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THE UNIVERSITY OF WARWICK

Optimization of last-mile delivery using Big Data and SAP ERP System

by

Priya Rithadia

Dissertation submitted in partial fulfilment for the Degree of Master of
Science in Engineering Business Management

WMG
University of Warwick

Submitted September 2025

Declaration

I have read and understood the rules on cheating, plagiarism and appropriate referencing outlined in my handbook and I declare that the work contained in this assignment is my own, unless otherwise acknowledged.

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The project should normally be related to the management of:

1. Companies in the engineering, retail and information technology sector,
2. The engineering function within a non-engineering company or
3. The supply chain within the engineering sector i.e logistics, procurement and sourcing.

The project may cover a wide range of topics, including operational, financial, human resource, technological, and strategic management challenges.

If the initiative is technological in nature, there must be unmistakable proof that the technology has a positive commercial impact.

The project MUST include a thorough comparative investigation of engineering sector practices if its concentration is outside of the aforementioned industrial range.

My project relates to this definition in the following way:

This study uses real-time data analytics and SAP ABAP simulation to improve supply chain operations in the engineering and retail industries. It particularly tackles a significant logistical challenge: last-mile delivery optimisation. Utilising SAP ABAP, the project develops a simulation model that enhances delivery time, reduces costs, and maximises truck utilisation by combining predicted demand, return data, and peak delivery insights. This model provides immediate business benefits by demonstrating how supply chain planning can be made more innovative and responsive with the aid of enterprise technologies, such as SAP. The project strongly aligns with the technical and strategic objectives of the engineering industry by demonstrating how simulation technology can provide quantifiable business advantages and clearly aid operational management in supply chains connected to engineering.

Abstract

Delivering products in busy city areas, especially in developing countries, is one of the most complex challenges in modern logistics. Roads are crowded, delivery demands change quickly, and existing tools often fail to handle local problems like poor infrastructure, traffic delays, and bad data.

This study explores how last-mile delivery (LMD) operations in emerging markets can be improved using data-driven simulation. The research focused on assessing the use of real-time data and demand forecasting, identifying the limitations in existing simulation models, and building a simple simulation using SAP ABAP. Publicly available data was processed and cleaned using Python to extract important information such as forecasted demand, peak days, and return flags. This cleaned data was then used in an ABAP-based simulation model. The model calculated key logistics performance indicators such as delivery time, cost, and number of vehicles required. These values were adjusted based on whether the day was a peak day or if returns were expected, helping to reflect more realistic delivery conditions.

The results show how combining real-time insights with simulation can help in better planning and resource allocation for last-mile deliveries. The research also highlighted the importance of data quality and the challenges in using real-world systems like SAP in academic settings. Overall, this study provides useful insights for improving delivery performance in emerging market contexts using accessible tools and simulation logic.

Key Words: last-mile delivery, big data, ERP, SAP, ABAP, forecasting, Python

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1. Introduction

In the fast-changing area of urban logistics, the efficient and timely delivery of goods to the final consumer, commonly termed the "*last-mile*" problem, has emerged as a significant issue, particularly in the case of emerging economies. This dissertation explores how the integration of real-time big data analytics with data-driven demand forecasting can significantly enhance last-mile delivery efficiency in urban retail supply chains for nonperishable goods using SAP ABAP.

The final leg of the supply chain, commonly referred to as **last-mile delivery**, distributes products from distribution centers to customers' homes. This phase has a high impact on customer satisfaction, delivery timeliness, and eco-green, although it spans a relatively small geographical region and usually represents more than 50% of the total logistics expense ("OPTIMIZING FIRST MILE AND LAST MILE OPERATION," 2025; Tran & Gavade, 2025). This inefficiency experienced by logistics is caused by fluctuating traffic conditions, heterogeneous infrastructure, and the challenge of navigating congested urban cores. The final mile of delivery is not only a logistical challenge but also a strategic opportunity for innovation and enhancement for emerging economies, which are further held back by challenges of fragmented urban planning, heterogeneous customer demand, and underdeveloped digital infrastructure (Freitag et al., 2021).

Big data analysis in the logistics sector involves deriving useful insights from huge volumes of both structured and unstructured data such as GPS logs, traffic patterns, consumer patterns, and cost. In order to facilitate dynamic routing, real-time demand forecasting, and timely decision making, real-time analytics emphasizes the immediate gathering and assessment of the aforementioned information (Freitag et al., 2021).

Demand forecasting is a methodological process that employs historical data, trends, and algorithms to predict future consumer demand. By leveraging real-time information to enhance forecast accuracy and responsiveness, retailers and logistic providers can modify schedules and capacities in response to fluctuating demand (González-Reséndiz et al., 2018a). Such integration can attain measurable operational efficiency and service reliability gains, even in urban retail contexts, for example, in emerging economies with unreliable infrastructure and heterogeneous data quality.

SAP is an abbreviation of Systems, Applications, and Products in Data Processing. It is a program that assists companies to carry out various processes-such as finance, logistic, and human resource- on one

platform. (Schenk & Draijer, 2019; Soosaimuthu, 2022) explain that SAP was designed to enable the efficient integration of business processes through the use of real-time data that enables more effective coordination across departments and faster decision-making.

Enterprise Resource Planning (ERP) is a software package that helps companies coordinate various aspects of their business within a single system. This involves functions such as finance, human resources, production and sales. It integrates all the departments with a unified database, as well as a similar procedure, enhancing the correlation and effectiveness of the business (Schenk & Draijer, 2019). Overall, SAP ERP can help companies automate business processes, integrate all business divisions and have real-time access to information and therefore, it is particularly valuable in fields such as logistics and supply chain management.

Unpredictable demand, logistical dispersion, and inadequate infrastructure are common problems in emerging economies (V. Silva et al., 2024). Reductions in costs, pollution, and service dependability are all possible results of applying data-driven models in these situations. Moreover, including environmental factors such as weather, as investigated in several research, offers an additional perspective for comprehending and enhancing performance in the face of real-world uncertainty (Mustafee et al., 2021).

This dissertation will utilize a simulation model that takes in dataset's inputs to demonstrate their impact on fleet performance, costs, delivery times, and demand fluctuations. The simulation will contrast dynamic, data-driven operations with traditional static models across various scenarios, including settings with high versus low data quality and peak versus off-peak times.

1.1 Background Study

A shift in the organizational and operating model of last-mile delivery (LMD) is observed, particularly in urban retailing within the emerging countries. This industry used to depend on heuristic routing and static planning, but it is now shifting to data-rich, adaptable systems that utilize real-time analytics. More specifically, the ability to interpret big data, incorporate prediction models and adjust to dynamic urban conditions such as traffic, customer volatility, and infrastructure inefficiencies are all the aspects of this transition (Freitag et al., 2021). Theoretical contributions in this discipline are broadly used in operations management and design of the cyber-physical systems. An example is the study that provides systems theory lens to the analysis of last-mile delivery, showing that the hybrid delivery fleet, which involves cars that are crowdsourced, requires decision-making structures that consider logistical demand patterns

coupled with real-time information (Castillo et al., 2022). Such systems perspectives help shift the focus of attention to the overall performance rather than pay attention to discrete performance measures.

1.1.1 Organizational Capability and Analytics Maturity

The concept of developing organizational skills considering occurrences in supply chains considers strategic alignment of data infrastructure in combination with decision-making frameworks and technical capacity and knowledge of analytics tools. Even under the assumption that the technological infrastructure is present, (Shrivastava & Riaz, 2023) emphasize that, due to their uneven IT skills and a poor analytics culture, SMEs in developing nations (such as the UAE) often find it hard to utilize AI and big data. This awareness leads to a fundamental problem of urban logistics: as long as companies do not include real-time information into their working and planning processes, it is no help.

1.1.2 Theoretical Grounding: Demand Forecasting and Last-Mile Delivery (LMD) Simulation

Study based on simulation has both advanced as a method of theory and method of simulating complex urban delivery environments. It allows experimenting with such factors as demand spikes, weather conditions, and routing algorithms without risking an operational breakdown. The simulations of drone delivery with the delivery life cycle modeling process models used to demonstrate the ability to outperform traditional approaches to transportation scheduling in situations of city logistics with constraints (Swanson, 2019a). These simulation models are aligned to the socio-technical systems theory since they offer platforms on operational innovation.

Some conceptual models and simulations presented in the reviewed sources are how delivery planning stands to benefit when coupled with BDA with data-driven demand forecasting. An example can be seen in the work of (Seyedan & Mafakheri, 2020) who simulate several forecasting models with the help of big data input and demonstrate dramatically better rates of prediction accuracy and cost-effectiveness in delivery environments. Therefore, their study to be an important theoretical basis of your simulation plan which is to replicate the delivery KPIs under your intervention and control condition.

Additional research demonstrates how these enhancements appear in important KPIs. The results pertinent to simulation model are summarized in the following table:

Author	Simulation Focus	Main topic covered
(Seyedan & Mafakheri, 2020a)	Forecasting models using big data	Validates BDA's effect on cost and time KPIs
(Huerta-Soto et al., 2023)	Pricing and supply chain agility	Shows integration of data forecasting and analytics improves agility in LMD
(Har et al., 2022)	Agent-based simulation of e-grocery LMD	Confirms scenario-based simulation validity

Table 1. KPI based on Simulation Model

1.1.3 Current Context: Urban Retail Supply Chains in Emerging Economies

In the emerging countries, last-mile delivery is costly and difficult to carry out logistically because of the unstable demand, poor infrastructure, and weather. Demand forecasting and big data analytics systems offer cushioning by transforming reactive functions of the system to predictive ecosystems.

Critical assessment of ERP systems that have integrated AI-based forecasting module is done. When the AI modules received real-time updates, the authors observed significant improvements in the supply chain visibility levels and a drop in the decision latency level (Fathima et al., 2024). Both the benefits of the study also note the dependence on the latency and quality of data, noting that real-time feeds are essential to simulation to capture actual limitations of the delivery. Research such as (Alves et al., 2023) and (Juan et al., n.d.) also evidences how scenario-based modelling, including such aspects as peak vs off-peak, or even bad weather, can enable accurate delivery planning. Simulation will provide a multidimensional way of analysing the overall impacts of changing demands and environmental disruptions using these approaches to last-mile KPIs, which are cost, competence in vehicle route management, and customer service support.

1.2 Research Problem

Despite the increasing volume of studies on last-mile delivery (LMD) innovation we know relatively little about how to combine data-driven demand forecasting (DDF) with real-time analysis of big data (BDA) to enhance delivery efficiency, particularly in the context of urban retail supply chains in emerging markets.

Incorporating BDA and AI with the tools of demand forecasting remains absent, even though these technologies have been widely explored to improve visibility and responsiveness throughout the logistics. Most LMD processes systems consider logistical implementation and data projections as distinct silos, limiting dynamic adaptation in rapidly changing cities, as (Masorgo et al., 2024) highlight. Likewise, developed urban logistics systems, too, remain insensitive to real-time conditions (Hadiansyah et al., n.d.), particularly in developing countries where unstable infrastructure is prominent. Some studies adopt simulation models to forecast LMD performance, but majority of the studies take static inputs and fail to consider key characteristics such as weather, peak hours traffic, and uncertainty in data quality. For instance, (Boom-Cárcamo et al., 2024) fail to consider any external disturbance in their dynamic model when estimating vehicle use, something that usually affects delivery efficiency, especially in developing cities.

What already know that real-time traffic data promotes route optimization (Al-Rahamneh et al., 2025) and AI-powered demand forecasting optimizes inventory and upstream logistics (Zhang et al., 2025). But in the literature on developing economies, there are very few studies that combine both layers forecasting and real-time response into a simulation architecture that mimics actual delivery situations. Relative to industrialized countries, emerging urban logistics systems are more fragile, dispersed, and data-poor, which exacerbates this imbalance. In spite of technologically feasible BDA technologies, (Lai et al., 2018) note successful deployment is stymied by organizational readiness, inadequate data infrastructure, and regulatory ambiguity. However, the below table 2 shows some research focused on simulation and associated gaps with it.

Study (Author, Year)	Simulation Focus	Gap Identified
(Boom-Cárcamo et al., 2024)	Vehicle utilisation modelling	No weather or real-time inputs
(Hadiansyah et al., n.d.)	Urban logistics platforms	Lacks peak/off-peak scenario modelling
(Masorgo et al., 2024)	LMD framework development	Siloed analytics and forecasting
(Al-Rahamneh et al., 2025)	Smart city logistics data use	Ignores delivery KPIs and forecasting link
(Lai et al., 2018)	BDA adoption in logistics	No application in emerging cities

Table 2. Simulation and Gap Identification

1.3 Research Question

The aim of this research is to extensively examine the beneficiary effects arising from the synthesis of real-time data streams namely, traffic, and order flow with predictive forecasting systems, through the combined use of Systematic Literature Review (SLR) and simulation. Apart from detailing the roles of the integration, the objective is to identify the most preferable conditions in which it can be implemented, particularly in urban domains of developing countries where logistic operations are being impeded by high volatility, poor infrastructure, and variable quality of information.

“How can real-time big data analytics integrate with data-driven demand forecasting to improve last-mile delivery efficiency in urban retail supply chains using SAP in emerging economies?”

1.4 Research Objectives

1. To assess how real-time big data analytics and data-driven demand forecasting are being used (or underused) in urban last-mile delivery systems within emerging markets.
2. To categorize and identify the main drawbacks of the current last-mile delivery simulation models, particularly their inability to integrate real-time input, focus on KPIs, and adjust to uncertainty in the real world.
3. To develop a simulation model using SAP system that integrates real-time traffic and demand data to optimise logistics KPIs, including delivery time, cost, and vehicle usage.
4. To evaluate the performance of the SAP simulation under varied conditions, such as peak vs. off-peak hours and varying levels of data quality (high vs. low).
5. To provide strategic recommendations for logistics decision-makers in emerging economies on implementing integrated analytics and forecasting through customised ERP models.

1.5 Research Justification

In developing nations, enhancing last-mile delivery (LMD) has become an economic necessity rather than a theoretical concept. Current LMD systems are marred by volatile demand patterns, ineffective utilization, and inefficiencies that have far-reaching impacts on the entire supply chain (Masorgo et al., 2024). In emerging economies, these inefficiencies are further exaggerated by infrastructural bottlenecks, dispersed data ecosystems, and congestion (Transforming Urban Logistics: Sustainable and Efficient Last-Mile Delivery in Cities, 2024a). A significant number of systems do not connect analytics with real-time operational execution, even with increased expenditures on big data analytics (BDA). Various research reflects that there exists a large discrepancy between the agility of decision-making and predictive analytics competencies (Raj et al., 2023). Though BDA technologies are utilized, they are not coupled with integrated logistics models or demand forecasting systems, especially in volatile retail contexts. The shortage is particularly severe in emerging economies because of the continuing evolution of retail dynamics and the failure of conventional systems to change accordingly (Castillo et al., 2022).

Operational benefits like increased route reliability, economic effectiveness, and delivery accuracy are potential results of combining real-time BDA with demand forecasting models, according to simulation studies (Swanson, 2019). Most of the current research is either limited to advanced economies or is theoretical and does not provide empirical support for developing scenarios. Significantly, simulation enables logistics tactics to be tried out in real urban settings, i.e., congested traffic, adverse weather, or late information, without disrupting ongoing operations. This permits systems to be constructed and decisions to be based on evidence (Juan et al., n.d.). Developing economies are presented with the chance to eradicate historical inefficiencies by maximizing their emerging digital ecosystems, provided these connections are planned carefully from the beginning.

The significance of this is that it looks at an empirical analysis of the impact of demand forecasting and the BDA on delivery performance under simulated urban constraints, constructed, uniquely, to match the demand of growing markets, as opposed to merely considering the concept of digital integration empirically. This directly affects the industry adaptation, technological development, and legislative basis that seeks to modernize the urban retail logistics.

1.6 Research Structure

This dissertation is divided into nine chapters, each building toward the development and evaluation of a simulation-based approach for improving last-mile delivery (LMD) in emerging markets using Big Data analytics and SAP ABAP.

Chapter 1: Introduction presents the background, research problem, and justification for the study. It introduces the role of Big Data in urban logistics, defines the research objectives (ROs), and explains the scope and limitations of the work.

Chapter 2: Literature Review reviews past studies on LMD, Big Data analytics, demand forecasting, and logistics simulation. It highlights key gaps, such as the lack of integrated ERP-based simulations using real-time data in emerging economies. The chapter also justifies the use of SAP ABAP and Python for simulation and data analysis.

Chapter 3: Methodology describes the two-part approach: a systematic literature review (SLR) guided by the PRISMA framework, and the development of a simulation model. It includes data selection, thematic analysis, dataset preprocessing in Python, and simulation logic in SAP ABAP.

Chapter 4: Simulation Model and Framework explains how the SAP ABAP simulation was structured, including technical setup, input logic, KPIs used (e.g., cost, time, vehicle use), and validation framework. It demonstrates how simulation results accurately reflect actual dataset patterns and support informed decision-making in urban logistics planning.

Chapter 5: Key Findings This chapter presents the main results found during the study. It is organised based on the research objectives. For each objective, the chapter explains what was discovered using both the literature and the simulation data. Key issues such as the limited use of real-time data in last-mile delivery, problems in existing simulation models, and how SAP-based simulation can help improve delivery time, cost, and vehicle use are discussed. The chapter also includes clear tables and graphs to support these points. A short comparison between expected and actual performance helps show where improvements are needed in real delivery systems.

Chapter 6: Results and Analysis presents the outputs from Python-based data processing. These include time series forecasting, peak demand identification, and customer order patterns. The findings are used as input conditions for simulation, aligning with the research questions and objectives.

Chapter 7: Limitations outlines the study's boundaries, including limitations in data scope, simulation complexity, and the generalizability of the model. It will acknowledge the practical and technical constraints encountered during the project.

Chapter 8: Conclusion will summarise the key insights and findings of the research. It will reflect on the contributions made to the field and whether the research objectives were achieved through the methodology applied.

Chapter 9: Future Recommendations will suggest how this study can be extended or improved. This may include testing with live ERP systems, using more advanced forecasting models, or applying the simulation in real-world logistics settings in emerging markets.

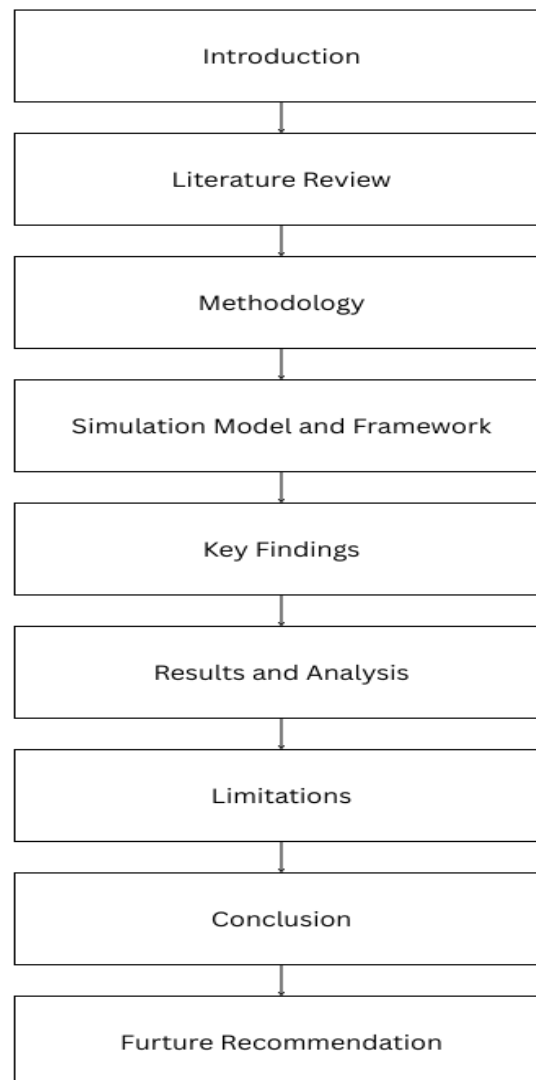


Fig 1. Research Structure

2. Literature Review

2.1 Understanding Last-Mile Delivery (LMD)

Last-mile delivery (LMD) is the final step in getting a product from a warehouse to the customer's door. Even though it's just a small part of the journey, it can cost up to half of all logistics expenses (Peppel et al., 2022). This is because urban areas face serious issues like traffic, pollution, and too many deliveries demands in too little space (Heikkinen, 2024).

In developing countries, it gets even harder. Roads are often poor, cities are unplanned, and data systems aren't strong enough to support smart delivery technologies (Sharma et al., 2025a). Even though there's lots of new tech like delivery drones or AI tracking, it doesn't always work well in these environments because the basic infrastructure is missing (Sorooshian et al., 2022).

Researchers have found that most past studies focus on either data or delivery, but not both together. That leaves a big gap, especially for solving real-life problems in developing cities (Masorgo et al., 2024). Clearly, LMD needs more than technology, it needs systems that understand people, roads, and cities (Zhu et al., 2023).

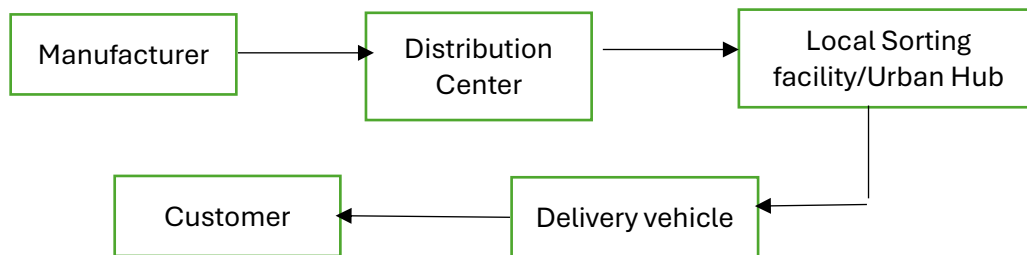


Fig 2. Block Diagram of LMD

2.2 Real-Time Big Data Analytics in Urban Logistics

Employment of IoT sensors, GPS, RFID, and AI/ML algorithms for processing high-speed logistics data for decision-making is the most important facilitator of real-time BDA for urban logistics (Fatorachian et al., 2025). Productivity in last-mile delivery (LMD) could be improved by utilizing these technologies, which comprise dynamic route planning, vehicle tracking, and scheduling in real time, especially in challenging metropolitan environments (He et al., 2022). Of interest is the fact that

while Big Data Analytics (BDA) has the potential to enhance Last Mile Delivery (LMD), especially regarding route flexibility and delivery time reduction (Judijanto & Rismanto, 2025), its implementation in developing nations remains scarce. Irrespective of the availability of infrastructural readiness, (B. N. Silva et al., 2018) argue that the leverage of real-time analytics is hindered by data integration and infrastructural problems. While Hadoop and Spark based BDA systems are possible in theory, (Rathore et al., 2018) further highlights this gap between theoretical possibility and real-world application by demonstrating that scalability and integration with sensor networks are challenging in data-fragmented urban settings.

The table 3. Below will discuss some implementation challenges and benefits of the BDA techniques used in urban last-mile delivery.

Study	Benefit Identified	Implementation Challenge
(Fatorachian et al., 2025)	Cybernetic feedback improves real-time adaptation	Infrastructure and interoperability
(He et al., 2022)	AI/ML aids micro-fulfilment in LMD	Lacks consistent data pipelines in cities
(B. N. Silva et al., 2018)	Real-time urban data aids planning	High cost and complexity of real-time systems
(Rathore et al., 2018)	Hadoop enables fast data processing	Difficult to deploy sensors at large scale

Table 3. BDA techniques used for urban LMD

There's also some disagreement about how practical BDA really is. For example, while (Fatorachian et al., 2025) show that combining IoT and predictive models can make logistics more adaptive, (B. N. Silva et al., 2018) point out that these systems often stay stuck at the prototype stage because cities in emerging economies don't have the real-time data infrastructure needed to fully support them. Likewise, (He et al., 2022) recognise the potential of AI to improve last mile delivery but admit that many developing countries still struggle with inconsistent technology adoption and weak policy support, making it hard to implement these solutions at scale.

2.3 Data-Driven Demand Forecasting and Its Integration with BDA

Forecasting demand has come a long way from old-school methods like moving averages and ARIMA to smarter tools like machine learning (ML), fuzzy logic, and hybrid models (Subramanian et al., 2025). Traditional forecasting methods mostly use past data and assume that demand stays steady over time. This makes them less useful in fast-changing urban retail environments (Hofmann & Rutschmann, 2018) also created a framework showing how big data analytics (BDA) could be used in forecasting at different decision levels daily, weekly, and strategic. But they also note that for this to work, countries need strong tech systems and skilled people, which many emerging economies still lack. On the other hand, (Subramanian et al., 2025) show that when BDA is combined with fuzzy logic (in their FL-BDDF model), forecasting becomes much more accurate, even when there's uncertainty like marketing campaigns or sudden demand spikes. This kind of model is ideal for simulating delivery systems that face constant changes. However, not all researchers are as confident. (Seyedan & Mafakheri, 2020b) argue that while ML models like gradient boosting and random forests are powerful, they're not yet widely used in actual retail operations. This is mostly because they're hard to explain and don't always work well with real-time data streams. This backs up the need for your simulation, which tests how these systems behave when real-time forecasting is used.

Model Type	Real-Time Ready	Used in Emerging Economies	Good for Simulation
Time-Series (e.g. ARIMA)	No	Yes	Some limits
Machine Learning (RF, GBM)	Partly	No	Yes
Hybrid (FL-BDDF)	Yes	Limited	Yes

Table 4. Demand Forecasting Models and Real-Time BDA Integration

There are some differences in how researchers see this topic. (Hofmann & Rutschmann, 2018) are hopeful about BDA's potential, but (Subramanian et al., 2025) aren't convinced it's ready for real-time use yet. The author takes a middle ground they believe hybrid models are very effective but still face problems when it comes to setting them up in places with fewer resources.

2.4 Simulation Approaches in Logistics

In logistics, simulation is used to compare options, test system performance, and make decisions without influencing with actual operations. However, the type of model, the availability of reliable data, and whether the model has undergone appropriate verification all affect how beneficial simulation is in practice. To enhance the reliability of simulation in bulk distribution logistics, for example, (Guiguet & Pons, 2025) created a three-phase validation methodology. In situations where real-time system monitoring is not feasible, their approach is quite helpful. Stakeholders may use it to verify that the model is rationally constructed, supported by reliable data, and produces accurate outcomes. This is significant since many simulations are still in use without proper validation, giving their results practically unreliable. Similar to this, (Richter et al., 2024) studied multimodal logistics systems that included port, rail, and road transfers using simulation. Prior to deployment, their methodology assists logistics customers in testing various scheduling or infrastructure configurations in a virtual setting. A more thorough understanding of operations is provided by combining digital twins with agent-based simulation, both at the system and individual levels. (K. O. A. N. Silva et al., 2024) provided another applicable example, using simulation to analyse the changes in e-commerce delivery during the COVID-19 epidemic in Brazil. Their model helped in assessing how delivery lockers can reduce carrier strain and improve productivity in emergency situations.

Simulation Model	Authors	Use
Discrete-Event Simulation (DES)	(Guiguet & Pons, 2025)	Simulating bulk distribution logistics with a validation framework
Agent-Based Simulation (ABS)	(Richter et al., 2024; K. O. A. N. Silva et al., 2024)	Modeling behavior of transport agents in a multimodal chain with digital twins
Hybrid (ABS + DES)	(K. O. A. N. Silva et al., 2024)	Simulating both system flow and delivery agents during the pandemic
System Dynamics (SD) + Monte Carlo	(Rabe et al., 2021)	Long-term planning of parcel lockers under demand uncertainty

Table 5. Simulation models and Use

Simulation has some significant limitations, despite its excellent application in logistics. A significant issue is the absence of real-time data. (Richter et al., 2024) clarify that the success of their agent-based

simulation using digital twins depends on the availability of regularly updated data. However, in the majority of actual logistics systems, this type of real-time data is either absent or delayed, thus the simulation cannot accurately represent what is occurring in real time. Another major problem is that a lot of models don't accurately depict actual metropolitan settings. (Rabe et al., 2021) demonstrate that standard techniques for calculating distances, such as grid or straight-line lengths, in last-mile delivery simulations don't accurately reflect how city streets function in real life.

2.5 ERP-Based Simulation (SAP Focus)

Enterprise Resource Planning (ERP) systems play a central role in logistics by managing business processes such as procurement, inventory, and transportation within a single, integrated system. In particular, SAP ERP is widely adopted by logistics companies because it helps standardise and automate complex workflows. As (Angolia & Pagliari, 2018) explain, SAP allows for realistic, data-driven simulation of logistics operations using real enterprise structures, which improves student and practitioner understanding of end-to-end logistics flows. The Advanced Business Application Programming (ABAP) programming language is essential to SAP as it enables users to create unique extensions and simulations within the ERP system. ABAP is particularly helpful for modeling logistics behavior using real company data since it is not only versatile but also closely coupled with SAP's business data and operations (Taratukhin et al., 2023a). Though these applications tend to concentrate more on learning ERP processes than replicating actual logistical dynamics, SAP's simulation tools are commonly utilized in education (e.g., classroom business games) and industrial scenario planning (Angolia & Pagliari, 2018; Taratukhin et al., 2023a). This creates a clear research gap. While logistics simulation is usually carried out using general-purpose tools like AnyLogic, Simul8, or Python-based platforms, few if any academic studies use ABAP to simulate LMD operations inside the ERP system itself.

2.6 Last-Mile Delivery (LMD) Challenges and Innovations

Last-mile urban retail delivery is a problem. Especially in new markets where there are many small shops. It is affected by traffic, scattered delivery providers, and the changing needs of customers. In most of the delivery solutions, the unreliability of the informal delivery chains is not taken into consideration because there is too much focus on the infrastructure (Mohammad et al., 2023). Even smart systems face real problems like unexpected drop-off points, diverse vehicles, and fluctuating

driver behavior (Sorooshian et al., 2022). Even though all these problems are known to everybody, the majority of studies still don't make enough use of real-time data. Robots and AI help with routing and efficiency, but most systems don't have real-time demand forecasting. This makes them slow to respond to spiky demand increases (Tsou, n.d.). (Ieva et al., 2025) propose a more advanced approach founded on rich data ecosystems, which are unavailable in developing countries. They use AI, virtual reality, and large language models to manage fleets with flexibility. (Alqaisi, n.d.) presents a blockchain-based decentralized delivery system that enhances trust and coordination, which is great but still lacks predictive analytics and is thus still not able to respond swiftly to abrupt demand shifts.

The real concern is that most of these models presuppose steady demand even while delivering earlier and at a lower price. Do not consider the volatility of demand, for example, temporary pop-up stores or peak hour rush, which are common in cities with unauthorised retail (Sorooshian et al., 2022).

2.7 Role of SAP ABAP in Logistics

SAP ABAP (Advanced Business Application Programming) is a programming language that is utilized for developing custom applications on top of SAP systems, particularly in the supply chain, logistics, and procurement modules. It enables the creation of data models, object-oriented business logic, and user interfaces that are tailored to individual operational needs (Zaidi, 2019). ABAP can be used to achieve the efficiency of data treatment, real-time processing of events, and smooth communication between transportation systems and delivery staff and customers by promoting efficiency, visibility, and satisfaction of customers in the last-mile delivery context (Kappauf et al., 2012). Its true power in logistics lies in the capacity to process real-time streams of data and facilitate automation of processes within key modules such as Extended Warehouse Management (EWM) and Transportation Management (TM). Through SAP TM, for instance, ABAP facilitates freight planning, carrier selection optimization, and shipment lifecycle automation. In order to create shipping activity, process transport data, and minimize downtime all of which are needed for last-mile delivery the system uses algorithms coded in ABAP (Ballamudi, n.d.). Its ability to customize reports and integrate with APIs further makes it useful in supply chain scenarios that demand accuracy and agility (Mangal, 2022). SAP ABAP chosen for this project because it lets us build our own custom simulation that fits exactly what we need. It can work with real-time data and allows us to create our own logic to improve delivery performance. Unlike ready-made logistics software, ABAP gives us full control to design a solution that matches the real problems in developing cities like unpredictable customer demand and poor

infrastructure. This flexibility makes it a perfect fit for solving last-mile delivery challenges in places where off-the-shelf tools often fail (Logistic Core Operations with SAP, n.d.).

2.8 Research Gap

Many studies talk about using simulations or forecasting in logistics, but they rarely connect these two in a practical, real-time way. For example, (Badakhshan et al., 2024) point out that combining machine learning with simulation can improve supply chains but most models don't use real-time feedback or live forecasting, especially in transport and delivery. Another problem is that real-world delivery issues like traffic, late orders, or missing data are often ignored. (Gürcan et al., 2025) suggest a smart digital twin model for delivery, but even that doesn't fully include things like messy infrastructure or sudden demand spikes. Similarly, (González-Reséndiz et al., 2018b) focus on lean logistics simulation, but they don't deal with unpredictable conditions like rush hour or roadblocks.

One major gap is the lack of flexible tools that can adapt to these problems especially in cities of developing countries. SAP ABAP, a language used to build smart logistics systems inside SAP, could help. But it's rarely mentioned in the research. (Angolia & Pagliari, 2018) show SAP how can simulate supply chain problems in education, but not for actual city delivery. Even studies about Logistics 4.0 (*2015 Winter Simulation Conference (WSC) : Date, 6-9 Dec. 2015, 2015*) or digital logistics systems (IEEE Staff, 2011) mostly talk about big-picture solutions not the real delivery struggles that happen on the ground.

This research fills that gap by using SAP ABAP to build a live, customizable simulation model that handles both real-time data and forecasting designed for the real problems that last-mile delivery faces in developing cities.

3. Methodology

3.1 Research Approach

The study uses two research methods, a Systematic Literature Review (SLR) and a simulation built using SAP software. These two methods work together, one gives a strong understanding of existing knowledge, and the other tests ideas in a realistic setting. The SLR was done in a structured and careful way, based on guidelines by (Cabrera et al., 2023). The goal was not just to collect articles, but to understand the state of research in ERP-based simulation and education. This approach helps find what is already known, where researchers disagree, and what areas need more study. To avoid missing important papers, the search was done using several databases like Scopus, Springer, Science direct, IEEE, Research Gate and Google Scholar. Using more than one database makes the review more reliable and less biased. This follows advice from (Foltýnek et al., 2020), who said that depending on just one source is risky. Only studies that were clearly relevant, high-quality, and related to ERP or simulation in learning were included. The simulation part of the research used SAP ERP software to create a business environment. This allowed students (or users) to actively take part in decision-making, not just learn through books. The simulations help learners better understand logistics and supply chain concepts by practicing them in a realistic system (Angolia & Pagliari, 2018). (Kramer et al., 2008) also support this type of learning. They used a simulation to help students work with accounting and business data, so they could see how their decisions affect company performance. (Draijer, n.d.) agree that simulations should use real-world tools like SAP, because this gives students skills that employers value. Using both methods together SLR and simulation makes the research stronger.

In addition to the Systematic Literature Review and SAP ABAP simulation, this research also includes a practical data-driven component. A publicly available “Retail Supply Chain Sales Dataset” was used. This dataset includes order-level details like Order Date, Ship Date, Sales, Profit, Country, Returned Status, and Customer ID. It was selected to reflect real-world supply chain flows and support forecasting, delivery pattern detection, and reverse logistics. Python (in Jupyter Notebook) was used to clean, prepare, and analyse the dataset. The outputs, such as order volume forecasts and customer segmentation, were used as input parameters for the SAP ABAP simulation model. The review helps build a solid base of knowledge, and the simulation allows testing that knowledge in action. This combination improves understanding and adds new value to both teaching and research in ERP-based business systems.

3.2 Systematic Literature Review (SLR)

3.2.1 Review Protocol and Eligibility Criteria

The PRISMA 2020 standards (Page, McKenzie, et al., 2021) were used in conducting and reporting this systematic review of the literature. Although not a methodological tool but a reporting guideline, PRISMA provided a clear framework for documenting the review process. It made it clearer how studies were identified, selected, and included (Sarkis-Onofre et al., 2021). A basic protocol for the search and selection process was established before the start of the review. It was followed to at all phases, although it was not registered in a public registry. By modifying the inclusion criteria as a reaction to research results, this internal protocol assisted in reducing the potential for bias (Page, Moher, et al., 2021). The research focused on studies that replicate learning through SAP or other ERP systems in supply chain, logistics, and decision-making settings. To ensure precision during the screening process, a clear set of inclusion and exclusion criteria is created as shown in table 4.

Criteria	Included Studies	Excluded Studies
Topic	Use of SAP or ERP systems for simulation-based learning or training, ERP/Big Data in logistics	ERP-focused studies without any simulation or educational application
Purpose	Aimed at improving learning, decision-making, or understanding of business processes	Focused only on system implementation or technical ERP configuration
Subject Area	Logistics, Supply chain, last-mile delivery, Big Data Analytics, Big data Techniques, demand forecasting, Simulation in Supply-chain, SAP in Supply chain and logistics	Solely depend on technical aspect of big data or Business in supply-chain in other sectors
Language	English	Non-English language

Publication Type	Peer-reviewed journal articles, books or conference papers	Abstracts, blogs, and grey literature
Full Text	Full text available	Only abstract available
Accessibility	Full-text and open-text articles accessible through institutional subscriptions or open access	Articles with no full-text available
Time Range	Published between 2015 and 2025	Published before 2015

Table 6. Eligibility Criteria of SLR

The abstract and title were used to initially screen every item that had been in the database search. The texts were only analyzed in their entirety if they met the first screening. To ensure transparent reporting and prevent selective inclusion, the PRISMA flow approach was the only guide for these activities (Page, McKenzie, et al., 2021). The research was highly transparent and reduced the potential for reviewer bias by utilizing pre-defined criteria and a systematic review process. This approach also ensured that only the highest quality and most relevant material was considered.

3.2.2 Search Strategy

A detailed search was carried out using five databases, Scopus, Springer, IEEE Xplore, ScienceDirect, and Google Scholar to identify the most suitable research for this study. Because they contain both technical and scholarly research on ERP, SAP, big data, simulation, and logistics, these databases were selected. Keywords related to the research's primary focus were used to build a search string:

("last-mile delivery" OR "urban logistics") AND ("real-time data" OR "big data analytics") AND ("demand forecasting") AND ("simulation" OR "SAP ABAP") AND ("simulation" OR "SAP" OR "logistic") AND ("simulation" OR "SAP" OR "last-mile delivery") AND ("last-mile delivery" OR "retail" OR "simulation") AND ("ERP based logistics simulation")

3.2.3 Selection Process

Throughout the selection procedure, the study followed to the PRISMA 2020 principles to guarantee that only the most essential and excellent papers were included in this evaluation (Page, McKenzie, et al., 2021).

The selection process was carried out in three main steps:

1. **Title and Abstract Screening:** First, all the articles retrieved from the search were reviewed by their titles and abstracts. Irrelevant studies that clearly did not meet the eligibility criteria were excluded at this stage as mentioned in table 4 above.
2. **Full Text Review:** For studies that passed the initial screening, full texts were downloaded and reviewed carefully. At this step, inclusion and exclusion criteria were strictly applied to assess the studies' relevance, context, and quality. Reasons for exclusion were documented.
3. **Final Inclusion:** After full-text evaluation, only those studies that fully satisfied all criteria were included in the final review. This ensured that the synthesis was based on high-quality and relevant evidence.

The entire selection process was documented using a PRISMA flow diagram as shown below in fig 3.

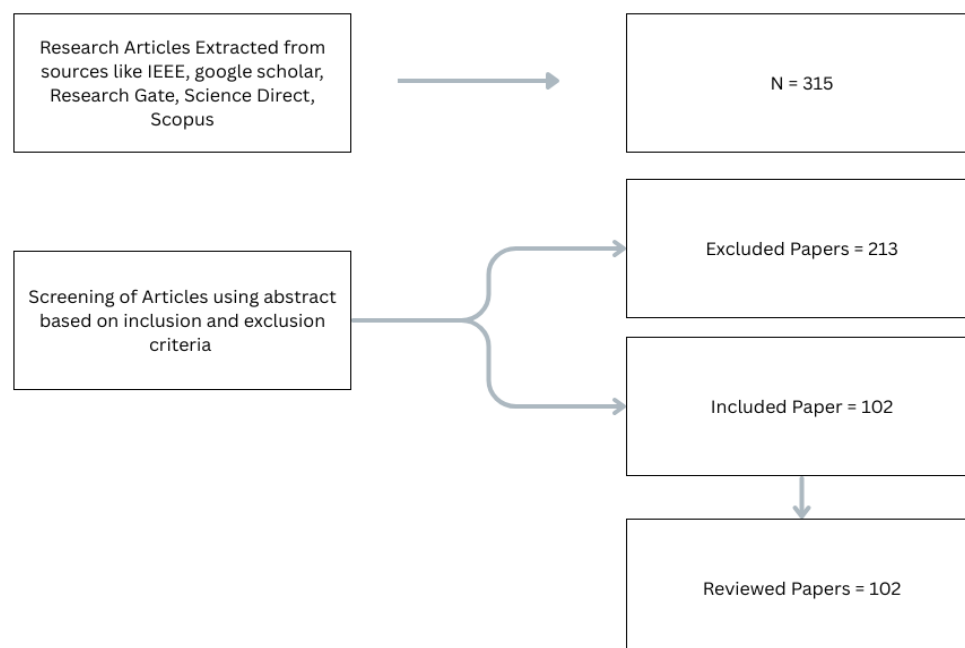


Fig 3. PRISMA Diagram

3.2.4 Data Extraction and Thematic Analysis

Each contained publication was understood by a standardized data extraction process to ensure standardization. This was followed by the active reporting of literature search and data management processes prioritized by the PRISMA-S extension (Rethlefsen et al., 2021). The description of such details was removed after reading through every paper critically:

Data Extracted	Purpose
Authors, year, and country	To understand trends over time and across regions
Methods and tools used	To identify use of ERP systems (especially SAP), simulation types, Big Data techniques used or models
KPIs and evaluation metrics	To observe what performance outcomes were measured (e.g., cost, delivery time)
Type of simulation/forecasting	To classify the study as using static, dynamic, predictive, or real-time systems
Gaps and limitations	To critically assess what was missing or poorly integrated in the model

Table 7. Data Extraction

Research Themes	Authors
Last-Mile Delivery Optimization and Technologies	(Agatz et al., 2025; Almasan et al., 2021; Barbosa et al., 2023; Beng, 2019; Castillo et al., 2022; Jaller et al., 2023; Janjevic & Winkenbach, 2020; Mohammad et al., 2023; Sharma et al., 2025b; Sorooshian et al., 2022b; Zhu et al., 2023)
Simulation and Modelling in Logistics	(Badakhshan et al., 2024; González-Reséndiz et al., 2018a; Juan et al., n.d.; Moon & Phatak, 2005; Mustafee et al., 2021; Rabe et al., 2021; Richter et al., 2024; K. O. A. N. Silva et al., 2024; Swanson, 2019a)
Big Data, AI, and Predictive Analytics in Supply Chain	(Eichenseer et al., 2025; Fathima et al., 2024; Hofmann & Rutschmann, 2018; Ieva et al., 2025; Lai et al., 2018; Raj et al., 2023; Seyedan & Mafakheri, 2020a; Shrivastava & Riaz, 2023; Subramanian et al., 2025)

SAP ERP and Supply Chain Systems	(Angolia & Pagliari, 2018; Ballamudi, n.d.; Banks-Grasedyck et al., n.d.; Chowdhury et al., 2025; Kappauf et al., 2012; Moon & Phatak, 2005; Ojha & Jaiswal, 2023; Soosaimuthu, 2022; Zaidi, 2019)
Sustainability and Smart City Logistics	(Al-Rahamneh et al., 2025; Fatorachian et al., 2025; Heikkinen, 2024a; Transforming Urban Logistics: Sustainable and Efficient Last-Mile Delivery in Cities, 2024a; Transport Forum at the OECD, n.d.)

Table 8. Thematic Analysis

3.2.5 Data Insight Generation Using Python

To better understand last-mile delivery behaviour, publicly available retail data was analysed using Python in Jupyter Notebook. This dataset included details about individual orders and deliveries. Key columns such as order date, ship date, return status, customer ID, and sales information were used to find useful patterns. These patterns helped in creating realistic simulation scenarios like demand forecasting, peak delivery days, late deliveries, and product return trends. Python was used to group, filter, and transform this data into usable formats. For example, daily order counts helped identify high-demand days, and return flags helped simulate reverse logistics. These results were exported into separate CSV files (such as forecast, return flag, and peak days) and used as input in the SAP ABAP simulation. Table 6 below shows the key variables taken from the dataset and how each one contributed to the project.

Variable	Description	Used For
Order Date	Date when the order was placed	Forecasting daily/monthly demand
Returned	Indicates if the product was returned	Simulating reverse logistics with return flags
Customer ID	Unique ID of the customer	Segmenting repeat customers or demand patterns

Table 9. Data Extracted from Retail Dataset and Use

3.3 Simulation Methodology

3.3.1 Simulation Design

The purpose of this simulation is to test how an ERP system handles common delivery problems like high order volumes, shipment delays, and returned products. It is a hypothetical example of practical implementation of SAP system which helps to understand how delivery operations might behave in real life, especially during busy or stressful situations. The simulation was built using SAP ABAP, the programming language used inside SAP ERP systems. Since many companies use SAP to manage deliveries and logistics, ABAP is a suitable tool for building and testing delivery processes in this study. A study justifies the application of simulation models in SAP ERP for training and analysing integrated business models, logistics, and real-time processes. For example, ERPsim games feature simulated real-world enterprise logistics systems, allowing users to gain experience in forecasting, procurement, and distribution within an SAP system. This (“REAL TIME DATA ANALYTICS AND DECISION MAKING IN ERP GAME SIMULATION: AN ACROSS DISCIPLINARY APPROACH,” 2018) corresponds with the proposed simulation strategy, as it is the best approach to simulate material handling, sales coverage, and demand circumstances in a relatively realistic and controlled environment. Moreover, discrete event simulation has also been used to extend the capabilities of ERP systems, producing detailed models of uncertainties and data relating to operations in real-time (Moon & Phatak, 2005). ABAP implementation of such simulations provides a close integration of their simulation with current processes of SAP and the data structures of the company, thus providing accurate and productive replication and testing of a complex delivery problem in the ERP system. The simulation represents a simple delivery process inside an ERP environment similar to as discussed approaches used in current studies. It includes:

1. Reading order data from a CSV file (created through Python analysis)
2. Identifying high-demand days based on order volume
3. Detecting returned products from the dataset
4. Flagging orders that may be delayed due to high demand

The logic was written using basic ABAP conditions. For example:

1. If the number of orders on a day is too high: mark some as delayed
2. If an item is marked as returned: move to return processing

The simulation uses real data from a retail supply chain dataset. This data was first analysed in Python to understand patterns and then converted into CSV files that were used as input in the ABAP program. A breakdown of overall simulation elements explained in below table 10.

Simulation Element	Source	Use in Simulation
Forecasted order volume	Python (time series)	Test high-load delivery scenarios
Peak sales days	Python (order volume analysis)	Simulate ERP stress days
customer list	Python (customer frequency)	Trigger priority delivery in SAP ABAP
Return order pattern	Python (returned flag)	Random Model reverse logistics
Simulation output	SAP ABAP logic	Delivery status, delays, return performance

Table 10. Inputs and Outputs of Simulation

Inventory is not part of this simulation. It is assumed that all products are available in stock, and the focus is only on delivery performance and order handling. This helps isolate and test delivery-related challenges without including warehouse or stock management. This simulation is not a full ERP system but a small model that helps study how delivery processes respond to real data. It focuses on delay handling and returns order management, using simple rules that can be found in real-world systems.

3.3.2 Input Data

The input data used for the simulation came from a real-world retail supply chain dataset in Excel format. This dataset contained detailed records of customer orders, including order dates, return status, quantities, and other related fields. For this study, only the fields that directly affect delivery operations were used. These included:

1. Order Date: to identify demand trends and busy days
2. Product Quantity: to measure delivery volume

3. Return Flag: to check how many orders were returned
4. Customer ID (optional): used for frequency analysis, but not simulated directly

The dataset was first cleaned and analyzed in Python using Jupyter Notebook. Python was used to:

1. Detect peak order days (high demand)
2. Track return rates
3. Forecast short-term demand using basic models like exponential smoothing

After processing, the relevant data was exported as CSV files, which were then used as input in the SAP ABAP simulation. The CSV files provided structured, real data that the ABAP program could read to simulate delivery behavior such as delays and returns.

3.3.3 Scenario Design

The simulation is designed to test how a simple delivery process behaves under two common situations based on real-world data: peak order days and returned orders. These scenarios were created using actual patterns from the dataset, which were identified and processed using Python.

Scenario 1: Peak Order Days

In Python, the dataset was grouped by order date to count daily order volume. The top 10 days with the highest number of orders were selected and saved into a CSV file. These dates were treated as high-load days in the ABAP simulation. During simulation, if an order date matches one of these peak days, the system tags that order as “delayed” due to high delivery pressure.

Scenario 2: Returned Orders

The dataset includes a column indicating whether an item was returned. This information was also prepared in Python and saved as part of the input file. In the SAP ABAP simulation, if the return flag is detected, the order is redirected to a return handling path, separate from normal delivery. These two scenarios were implemented in SAP ABAP using basic conditional logic. The program checks each order record from the input CSV and decides how to process it:

1. If the order date is on a high-demand day: mark as delayed
2. If the return flag is active: mark as returned
3. Otherwise: process as a normal delivery

3.3.4 Output Matrix

The output of the simulation represents how the system handled each order based on predefined conditions. After reading the input CSV file, the SAP ABAP program classifies each order into one of the following output types:

1. Normal Delivery: Orders that were not returned and did not occur on peak-demand days.
2. Delayed Delivery: Orders placed on peak-demand days, where simulated delivery capacity was assumed to be under pressure.
3. Returned Order: Orders marked with a return flag, simulated to be handled through a reverse process.

Each simulated order is assigned to a status field in the output table, based on the logic implemented in the ABAP code. The results are printed on the SAP console and can optionally be exported into a flat file (CSV or internal table).

The output helps in understanding:

1. How many orders were delayed due to simulated high demand?
2. How many were returned?
3. How many were processed normally?

This output was also compared against the original Python analysis to validate whether the simulation reflected the expected behavior of delivery flow under realistic conditions.

3.3.5 Validation

Since this simulation was developed in SAP ABAP and based on custom logic, the validation was done through internal checks rather than external benchmarking. The goal of validation was to confirm that the simulation behaves logically and processes the input data correctly. (Moon & Phatak, 2005) paper also outlines the necessity of verifying simulation models through real-world system performance data, along with critical performance indicators (KPIs). It focuses on iterative verification and validation through scenario testing and stakeholder feedback in the ERP setting. This is because the simulation is embedded with SAP ERP through ABAP (R/3 Architecture). Validation also proves complementary between SAP transaction data, system logs, and simulation outputs to ensure the accuracy and reliability of the modelled delivery processes.

The following steps were used to validate the simulation:

1. Code logic testing: Each condition (high-demand day, return flag) was tested individually with sample input data to ensure the ABAP program assigns the correct delivery status.
2. Controlled input cases: Small CSV files with known values were created to test how the system handles expected and edge cases. For example, dates known to be peak demand days were entered to check if delays were correctly flagged.
3. Output review: The output log was manually reviewed and compared against the input data to confirm that all orders were processed as per the defined logic.

No performance or time-based simulation metrics were used, as the focus of this project is on logical behavior rather than system performance or full ERP replication.

3.3 Article Proofing

In conducting the literature review for this study, 102 scholarly papers were identified and examined. The majority of the papers selected are from recent years, ensuring that they capture current events. The number of articles published between 2020 and 2025 totalled 65, indicating that most of the research applied in the current study is up-to-date and applicable. An additional 28 papers supporting this field were published between 2015 and 2019, highlighting the current state of knowledge and facts related to the research. A few older papers were included due to their exceptional contribution. Two articles published in 2005 and 2011 were included as they offered a unique insight; SAP ABAP did not receive much attention in newer literature. As the study of SAP ABAP in the logistics simulation environment is scarce in recent academic contexts, older works had to be used to support the study. Figure 4. represents the number of papers published based on SAP and logistics/last-mile delivery.

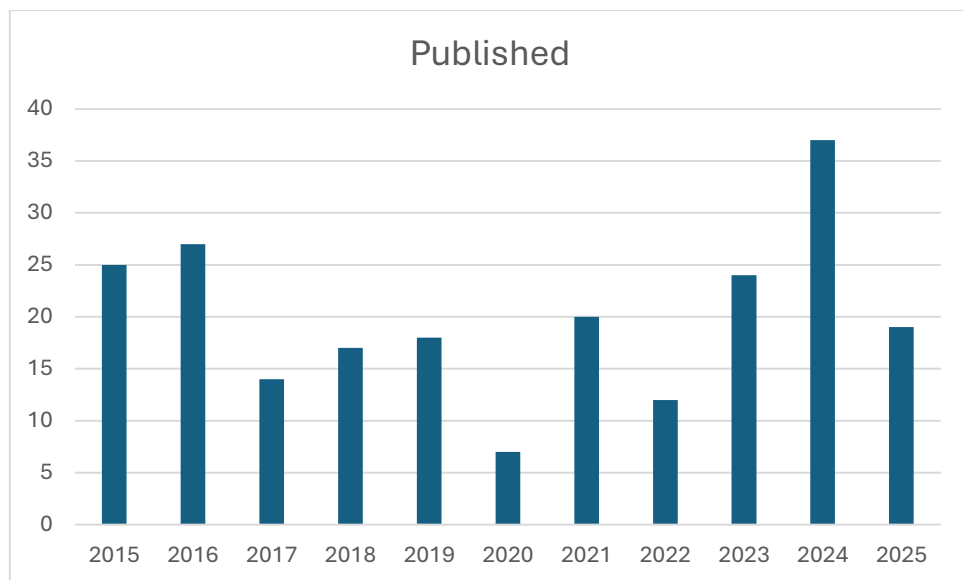


Fig 4. Published paper each year record

4.Simulation Model and Framework

This chapter describes the simulation model developed to evaluate the potential of data-driven demand forecasting and real-time big data analytics in enhancing last-mile delivery (LMD) in urban retail supply chains using SAP ABAP. The model directly addresses the study topic and the previously noted problems: most studies, particularly in emerging nations, do not integrate forecasting and real-time reaction within an ERP environment. Key outcomes, including delivery time, cost, and vehicle utilization, are measured by the simulation, which focuses on real-world delivery challenges like peak days, data quality, and returns.

4.1 Conceptual Framework

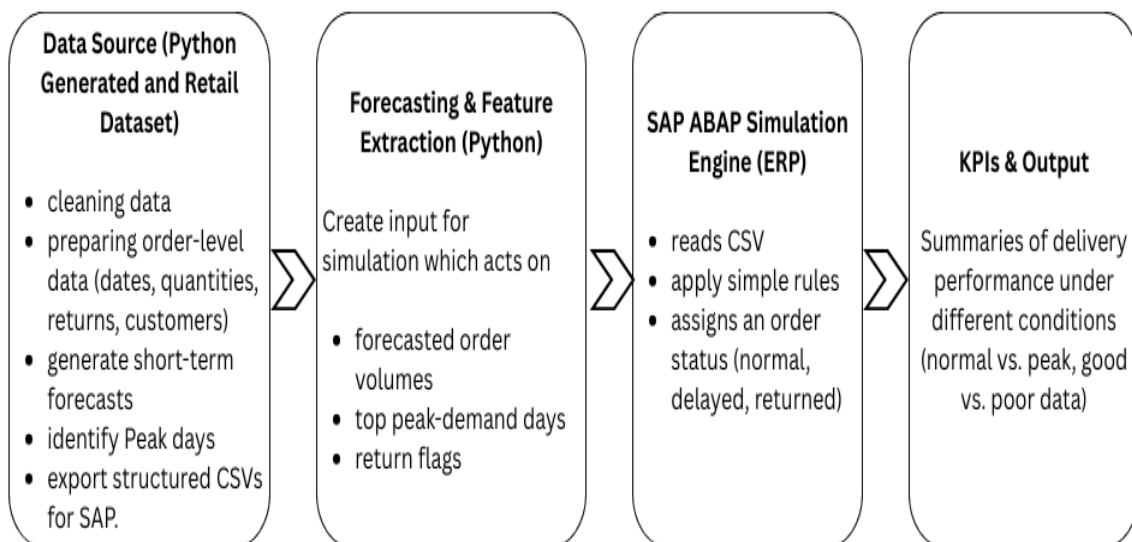


Fig 5. Conceptual Framework

The transformation of analytics results into inputs that the ERP can utilise is depicted in Figure 5. The pipeline works well and is simple to restart. Simple features are created for simulation, and orders are aggregated at the calendar-day level in Python. Returns are highlighted from the dataset, and peak days are selected using a top N selector that may be customised. For SAP to read these characteristics consistently, they are stored as flat, clean files. Intentionally, the hand-off is one-way. ABAP reads CSV files but does not alter the YYYY-MM-DD dates or fixed column names. Every record in SAP has a single, unchanging status: Normal, delayed (if the date is a peak day), or Returned (if the return flag is set). The mapping always produces the same outputs because it does not store any memory. Results are

generated at two levels: daily totals that can be compared across scenarios, and order-level labels. This facilitates quantifying the impact of stressful times. Because of the modular architecture, current rules remain functional while a new feature necessitates adding a column to the input. The model's boundary is the assignment of delivery status; routing, inventory, and optimisation are not included.

4.2 Simulation Environment

SAP ABAP Public Edition and Python in Jupyter Notebook were the two primary tools used during this study for analysis and simulation. Python was used because it offers flexible forecasting and data preprocessing capabilities. For the sales dataset, time series forecasting techniques, such as Simple Exponential Smoothing, were employed. Because it originated from a retail supply chain, the data was ideal for time-based order forecasting. Because Jupyter Notebook allows for step-by-step coding and visualization, it makes the process easier. This allowed the research to analyze and clean the data before the SAP simulation swiftly. Exporting the predicted values, peak sales days, and client order frequency for additional processing in SAP ABAP was made feasible by Python's proficiency with CSV files (978-1-4302-4804-0, n.d.; Copyright Introduction to Time Series Forecasting with Python, 2020). Moving these datasets into SAP ABAP, which was selected as SAP's primary development platform for business logic and simulation, was the next step. Since it offers a contemporary, cloud-based development environment with access to fundamental SAP features, such as internal tables, modular programming, and Open SQL, all of which are necessary for handling our CSV-based input, therefore, chose to utilise the Public Edition of ABAP (Grześkowiak et al., 2019). With SAP ABAP Public Edition, the Python datasets (forecast_for_abap.csv, peak_days.csv, and returns.csv) were imported using either the table maintenance interfaces or GUI_UPLOAD. The logic created to simulate delivery delays and customer segmentation scenarios was then applied to these files in ABAP applications. This hybrid simulation technique facilitates testing of delivery thresholds and understanding of high-value customer behavior, thereby supporting both operational and decision-making objectives.

4.3 Framework for Analysis

The performance of the simulation model is assessed using a structured analytical approach. Using Python-based forecasting, this system compares the original demand patterns with the outcomes of the SAP ABAP simulation. The objective is to investigate the effects of various delivery circumstances, such as high or low demand, on key logistical metrics, including cost, vehicle utilisation, and delivery time. Importing the predicted order volumes (10 days) produced in Python using Simple Exponential Smoothing is the first step in the study. Conditional logic is then employed to replicate real-world

logistics performance using these as direct inputs into the SAP simulation model. Based on business rules specified in ABAP logic, such as delivery times extending on busy days, the simulation handled each order day separately and computed KPIs. Comparing the order trends with the simulated KPIs is the last stage. For instance, it is assumed that increased order volumes would result in longer delivery times and more vehicles being used; the SAP outputs are used to evaluate this link. The framework ensures that the analysis accurately reflects the various demand situations (such as normal vs. peak days) and aligns with the realities of logistics in the real world. This system facilitates repeatability and offers traceability from raw data to the final simulation results. The model's ability to manage dynamic logistics scenarios can be verified by comparing simulated responses with actual data trends.

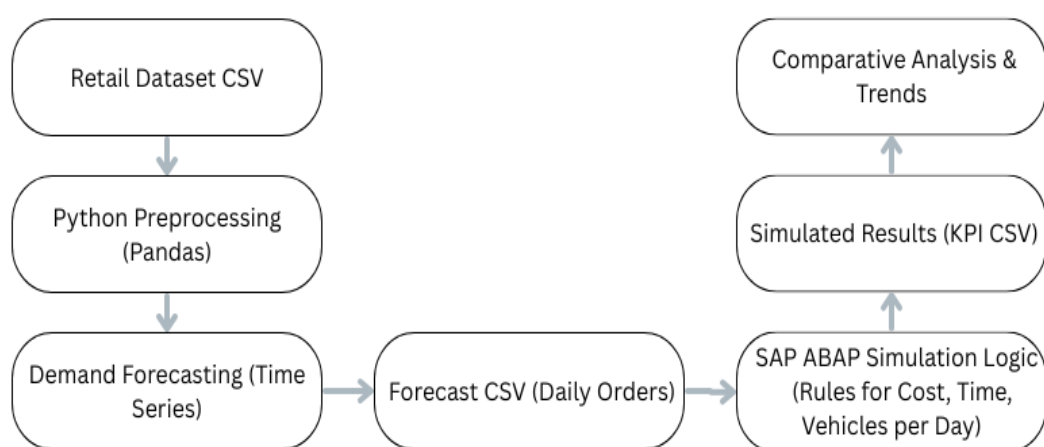


Fig 6. Analysis Framework Flow diagram

This structure supports RO3 and RO4, where simulation-based testing is necessary under variable demand conditions. The flow also reflects a data-driven decision-making pipeline often seen in ERP-integrated logistics planning, as supported by industry models (Angolia & Pagliari, 2018; Chowdhury et al., 2025).

4.4 Validation Approach

Instead of deploying simulation in the actual world, a logical testing method was used to validate it. The goal of this procedure was to verify that the simulation operates as anticipated when structured inputs and conditions derived from the cleaned Python dataset are used. Code-level testing using SAP ABAP was the initial step. To verify that conditional expanding and loops operated as intended, each simulation module was run using sample inputs (daily order quantities, return flags, and customer IDs). This approach adheres to the procedures outlined by (Angolia & Pagliari, 2018), who used ERP-based simulations to simulate logistics

processes in a controlled setting. The use of controlled test settings followed. To verify whether delivery time and cost increased rationally, for example, data for days with high demand were fed into the simulation. Similarly, to confirm that the model could reproduce routing or efficiency shifts, repeated customer orders were employed. These inputs originated from a previous analysis conducted using Python. Lastly, a comparison was made between the simulation outcomes and the patterns observed in the Python data, specifically from the peak demand analysis and time series projection. This made it possible to compare the results of the ABAP model with the predicted results (such as longer delivery times on busy days), guaranteeing behavioural consistency. Such simulation-based validations are frequently suitable for evaluating delivery models in logistics systems under imaginary stress situations, as (Chowdhury et al., 2025) point out.

5. Key Findings

5.1 RO1: The earlier parts of the Literature Review, particularly parts 2.2 (Last-Mile Delivery in Emerging Markets), 2.3 (Big Data Analytics in Urban Logistics), and 2.4 (Demand Forecasting in LMD), have already covered some of the material relevant to this objective. The applications of big data and forecasting tools in logistics were emphasized in these sections, along with the associated problems, particularly regarding ERP integration and real-time applications. The main conclusions for RO1 are presented below in the context of this discussion.

According to several studies, logistics companies can utilise big data to rapidly adjust to changes in weather, traffic, and order volume. Many emerging cities still have little integration of this data, however. For example, despite the availability of massive datasets from GPS, mobile phones, or sensors, logistics operations in places like São Paulo or Nairobi still rely on human planning or older routing algorithms (Eichenseer et al., 2025; Janjevic & Winkenbach, 2020). The majority of current implementations are found in developed markets with real-time integration enabled by infrastructure. Studies such as (Eichenseer et al., 2025), on the other hand, demonstrate that while data is frequently gathered in developing economies, it is not transformed into valuable insights. This results in lost chances for demand planning or route optimisation. Conventional techniques, such as seasonal models and ARIMA, are still often employed in forecasting. For short-term demand forecast in urban delivery, however, new studies have begun utilising ML and AI models (Barbosa et al., 2023; Bruni et al., 2023). Nevertheless, especially in environments with limited resources, these models are rarely connected with ERP systems such as SAP.

One of the main gaps identified is the lack of simulation environments that assess how real-time data and predictions impact delivery KPIs, such as cost and delivery time. To address this, the current dissertation employs time series models in Python, utilises a Kaggle dataset, and integrates the results into a SAP ABAP simulation environment. These results corroborate with RO1 by demonstrating that, although real-time BDA and forecasting have shown promise in research, their actual use in developing urban logistics systems remains scarce and poorly understood.

Reference	Key Finding Supporting RO1	Focus Area
(Janjevic & Winkenbach, 2020)	LMD in São Paulo still lacks real-time data integration	Urban data gap
(Eichenseer et al., 2025)	Emerging markets often collect but do not utilise data	Data underuse
(Eichenseer et al., 2025)	Big data aids dynamic logistics but adoption is uneven	Use of BDA
(Barbosa et al., 2023)	Demand forecasting using ML has potential in LMD	Forecasting
(Bruni et al., 2023)	AI/ML can improve short-term LMD prediction	Forecasting models
(Sharma et al., 2025b)	Simulation environments are rarely integrated with ERP in urban settings	Simulation gap
(Transport Forum at the OECD, n.d.)	BDA frameworks exist, but practical delivery applications are low	Practical use gap

Table 11. Supporting Reference for RO1

5.2 RO2: Several concerns with last-mile simulation models were previously identified in the literature review. Among these were difficulties in real-time responsiveness, poor KPI monitoring, and challenges adjusting to irregular delivery situations. Building on this, the results for RO2 utilise data from previous studies to examine these limits in greater detail. The fact that most last-mile simulation models are now based on static input data is a significant issue. (Jaller et al., 2023) demonstrate, for instance, that delivery models typically rely on pre-established delivery scenarios and do not adapt dynamically to real-world events, such as unexpected spikes in demand or traffic jams. This restricts its use in busy, uncertain metropolitan settings. Another common drawback is the lack of a clear KPI focus as per the study. While models may simulate delivery routes or times, they often ignore specific logistics performance indicators such as cost, vehicle efficiency, and delay penalties (Guerra-Regalado & Guevara, 2025). As a result, decision-makers do not get actionable insights that align with business priorities. Moreover, many models struggle to manage uncertainty. To manage unpredictable urban elements, such as diverse and uncertain real-world complexities, (Fahim et al., n.d.) noted that existing models do not incorporate live feedback loops from operational data. These models frequently use "ideal" assumptions, which reduces their applicability in the actual world. Additionally, simulation frameworks frequently lack flexibility and are not intended to be easily integrated with machine learning or big data technologies. They are therefore outdated when considering contemporary supply chain systems (Srai, n.d.). This dissertation's methodology, on the other hand, builds upon SAP ABAP simulation and Python forecasting, which, when combined, provide dynamic input processing, KPI tracking, and flexibility to handle peak load or return scenarios, problems that current models generally overlook. The model is now more in line with operational requirements, as these constraints have been directly addressed in the SAP logic by incorporating factors such as delivery cost, vehicle utilisation, and returns.

Reference	Main Focus
(Beng, 2019)	LMD relies heavily on optimization and simulation models to test new delivery initiatives, focusing mainly on efficiency metrics (cost, lead time, utilization, distance). Highlights the prevalent use of these models and their typical performance indicators.
(Beng, 2019)	Demonstrated the benefits of accounting for travel time variability in last-mile distribution. This underscores that many earlier models, which assumed fixed travel times, were deficient in handling uncertainty; incorporating variability leads to more robust delivery plans.

(Agatz et al., 2025)	Find that academic LMD models have a narrow, rationalist focus on resource allocation (e.g. routing costs), lacking a unified framework for the broad sociotechnical challenges of last-mile operations. They call for integrating perspectives like service operations and human factors to address real-time variability and multi-dimensional performance in last-mile systems.
(Fahim et al., n.d.)	Identify a gap in existing research: most studies focus on isolated challenges (routing or demand in silo) without integrating multiple factors. Developed an integrated model combining operational, financial, and environmental metrics, showing that prior models rarely captured such multi-faceted trade-offs. Confirms that traditional simulations often ignore sustainability and holistic KPIs.
(Sharma et al., 2025b)	Highlights that real-time tracking, AI-based dynamic routing, and digital twins are critical enablers for efficient LMD. Implies that legacy simulation models (without these enablers) lack the ability to adapt on-the-fly. Shows how I4.0 technologies overcome current models' rigidity by enabling continuous data integration and responsiveness in last-mile delivery.

Table 12. Supporting References for RO2

5.3 RO3: Some information mentioned in the earlier literature review already supports RO3. For instance, the literature discussed how SAP S/4HANA enables logistics processes through data visibility, integration, and automation. This includes the platform's capacity to support decision-making with real-time operational and experience data (O-Data and X-Data), as noted by (Banks-Grasedyck et al., n.d.) in *Successfully Managing S/4HANA Projects*. In addition, (Ojha & Jaiswal, 2023) explained how SAP S/4HANA's asset management and maintenance functions connect operational logistics with predictive insights. The recommendations that follow are based on the sources provided for RO3, which aims to create a simulation model using SAP that incorporates real-time traffic and demand data to optimise logistics KPIs (such as delivery time, cost, and vehicle utilisation). Integration of real-time data from external and internal systems is possible with SAP S/4HANA. The ability to make dynamic modifications to planning and scheduling makes it appropriate for logistics simulations (Banks-Grasedyck et al., n.d.). The ICID 2022 conference papers, for example, offer frameworks for modelling complex service delivery systems that may be modified for logistics optimisation. Simulation models are regularly debated in studies like this. Operations systems can benefit from simulation, as demonstrated by digital twins and real-time simulation techniques (Taratukhin et al., 2023b). These techniques provide pertinent approaches for predicting and responding to logistical disruptions based on real-time inputs. Furthermore, (Ojha & Jaiswal, 2023) highlight how SAP Asset Management simulates vehicle and resource workloads using work center capacity planning, which is closely linked to RO3 goals. SAP's real-time analytics and monitoring tools embedded in S/4HANA allow logistics managers to forecast KPIs such as cost and delivery time. SAP Fiori applications further support this by offering visual and user-friendly tools to reconfigure logistics planning dynamically (Ojha & Jaiswal, 2023). Moreover, the SAP Activate methodology, explained by (Banks-Grasedyck et al., n.d.), includes phases like Explore and Realize that support data modelling and simulation before deployment, making them useful for building logistics simulation workflows. Additionally, the Proceedings of the 20th SAP Academic Conference highlight the effectiveness of incorporating SAP Build.me and design thinking methodologies into classroom instruction, enabling students to co-create application prototypes in collaboration with stakeholders. This not only nurtures their application development skills but also fosters critical business analysis competencies (Bliemel et al., n.d.). Complementing these insights, (Super et al., 2020) explored the cognitive and motivational mechanisms underlying simulation game performance, revealing that knowledge-sharing norms and the development of transactive memory systems significantly impact students perceived team performance and satisfaction. These findings validate the integration of ERP-based simulations and design-led pedagogy as instrumental in cultivating both technical acumen and collaborative problem-solving abilities within enterprise system learning.

5.4 Impact on Sustainability

This research project involves optimising last-mile delivery (LMD) using simulation models developed in SAP ABAP. The strategy was tested in terms of sustainability, specifically in relation to environmental and operational aspects. Last-mile delivery is a significant contributor to urban challenges such as emissions, congestion, and safety hazards. The World Economic Forum (2024) confirmed this by stating that the increase in emissions resulting from urban deliveries may grow by 60 per cent by 2030 without the introduction of improved practices (Transforming Urban Logistics: Sustainable and Efficient Last-Mile Delivery in Cities, 2024b). That is why it is significant to employ effective and evidence-based planning tools. In this study, the SAP-based model facilitates real-time decision-making by simulating traffic conditions, order returns, and demand patterns. All of these are significant concerns that impact emissions, vehicle usage and delivery timing. (Heikkinen, 2024b) revealed that a significant subset of logistics managers continues to conceptualise sustainability as solely being roughly equivalent to the reduction of emissions, and tend to disconnect it from other valuable objectives such as worker well-being or economic performance. This project aimed to fill that gap by incorporating cost, time, and vehicle use into the simulation's logic, which can lead to more informed sustainability choices. The solution can assist companies to plan routes more effectively, minimize waste and use fewer vehicles due to predicting real demand and returns. Sharing data tools and infrastructure is the key to the establishment of a sustainable future of deliverier (Transforming Urban Logistics: Sustainable and Efficient Last-Mile Delivery in Cities, 2024b). Through simulation in SAP, this project favors that trend and is able to enable the companies to plan more sustainably, particularly in cities where demand is increasing.

6.Results and Analysis

RO4: This chapter presents the results obtained from two parts of the study: the big data analysis and the SAP ABAP-based simulation. The analysis focused on exploring order trends, forecasting demand, and identifying high-demand days. The insights generated from this dataset were then used as input for a simulation model developed in SAP ABAP. The purpose of this simulation was to evaluate how delivery-related performance indicators such as delivery time, cost efficiency, and vehicle utilization are affected under different conditions, including normal demand and peak load scenarios. These combined results provide a better understanding of how real-time data and forecasting techniques can enhance last-mile delivery efficiency, particularly in the context of urban logistics within emerging markets. For full source code, technical implementation details and csv files, please refer to the Appendix.

6.1 Big Data Analysis Results

6.1.1 Time Series Forecasting

To analyze short-term demand trends in the retail supply chain, a time series was developed by grouping customer orders by date. The dataset used for this analysis was an open-source retail supply chain dataset obtained from Kaggle (*Retail Supply Chain Sales Dataset*, n.d.). This dataset contains transaction-level data, including timestamps, customer identifiers, and order information. Daily order counts were aggregated from the Order Date field to create the time series input. The data was then pre-processed in Python to remove any missing or duplicate entries before being analyzed using time series forecasting techniques.

The Kaggle-based retail sales dataset was utilized as an example to categorize the number of orders by calendar date, enabling the construction of a time series to better understand daily demand fluctuations in the retail supply chain (*Retail Supply Chain Sales Dataset*, n.d.). In the context of an ERP system, this provided a foundation for analyzing short-term demand behavior, which is crucial for downstream modeling. To produce a 10-day forecast of future orders, Simple Exponential Smoothing (SES) was applied to the time series. SES was chosen due to its suitability for short-term demand forecasting in volatile retail environments, especially when the data is frequently updated and trends are not linear (H. Abdelati & Hilal A. Abdelwali, 2024; Kumar Paul

& Sarkar, n.d.). The model was developed using Python, and the forecast was presented alongside the last 100 days of actual order data (see Figure 7).

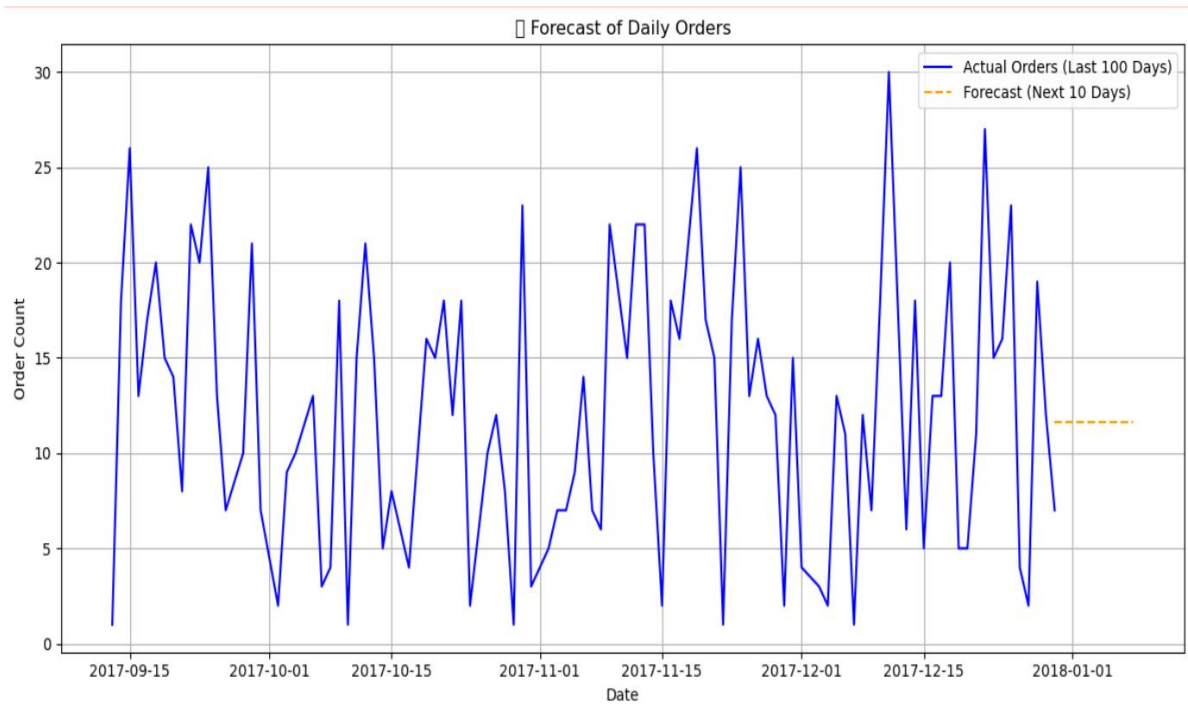


Fig 7. Daily Order Forecasting

With daily variations ranging from less than five to over twenty-five orders, the time series graph displays significant daily variations. The nature of urban retail demand is reflected in this high degree of fluctuation, which is influenced by several short-term factors, including client preferences, promotional activities, and external disruptions. On an average of 12 to 15 orders every day, the range predicted for the next 10 days looks relatively constant. This smoothed trend offers a realistic and practically manageable input for ERP-based simulation. Studies that highlight the practical utility of short-term demand modelling in logistics systems provide credence to the applicability of such a prediction. For example, (Li & Wei, 2022) demonstrated how lightweight statistical forecasting models, such as SES, can aid logistics control decisions in situations where comprehensive machine learning pipelines are ineffective or unavailable. Furthermore, (Fadda et al., 2021) emphasis how time series forecasting helps last-mile delivery settings match tactical planning with actual demand cycles. The forecast result is exported and used as input for the SAP ABAP-based simulation model, which is used to analyze logistics KPIs, including truck load efficiency and delivery schedule.

6.1.2 Peak Demand Days

To understand how the system performs during heavy workloads, the retail dataset was analysed to find the ten days with the highest number of customer orders. These days are illustrated in Figure 8, where a clear spike in order volume is visible. These high-demand days were selected because they represent times when delivery systems are under more pressure than usual.

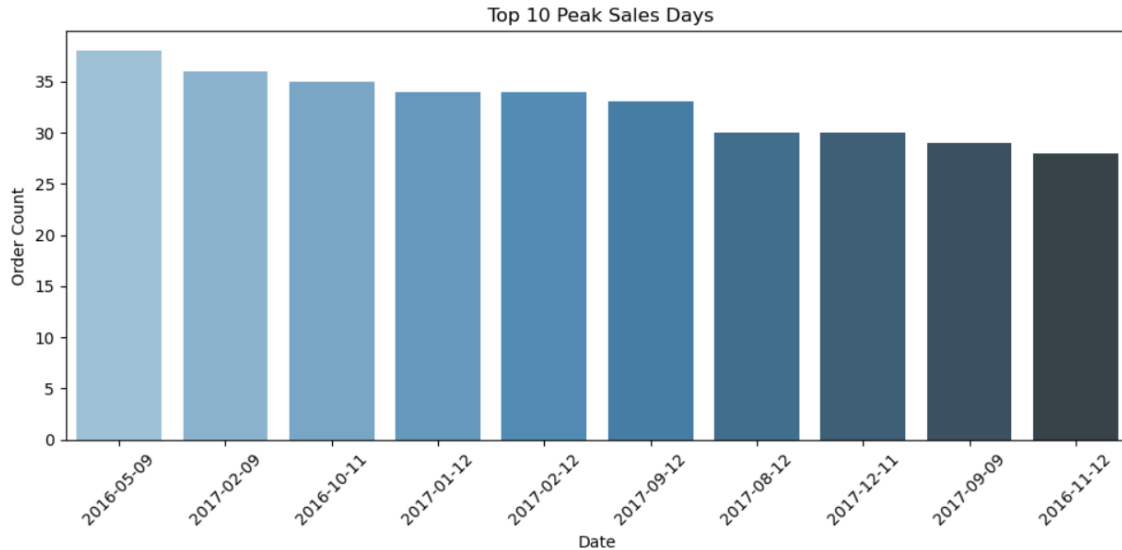


Fig 8. Peak Days Sales

Demand surges can lead to significant delays, increased costs, and poor delivery performance in urban supply chains, especially in developing countries (Chowdhury et al., 2025). During peak periods, delivery trucks may be overbooked, routes may become inefficient, and customer satisfaction may decline. These peak days are used in the SAP ABAP simulation to verify the system's ability to withstand pressure. In logistics models that rely on ERP, this type of stress testing is crucial. According to (Angolia & Pagliari, 2018) SAP simulations may be used to assess how well the system functions in instances with abnormally high demand by simulating genuine, high-load scenarios. The high-demand days will also allow better modeling of real-life situations. (Almasan et al., 2021) emphasize that contemporary logistic systems should be prepared to deal with shifts in demand and traffic. Incorporating such days into the simulation enables the system to be checked whether it can respond to these changes in real-time. This study used those peak demand days to replicate logistics stressful situations in SAP ABAP. This was used to determine whether the simulation was still able to output on some of its key performance parameters, delivery time, and cost of delivery, and use of vehicle when demand was at an unusually higher rate.

6.1.3 Customer Order Frequency

In this part, the retail dataset is used to analyse the frequency of consumer orders. Based on the frequency of their orders, the top ten most frequent customers were identified. Figure 9 displays the outcome, with each bar signifying the total number of orders placed by specific clients. This pattern identifies repeat clients who make frequent purchases. Due to their propensity for more consistent behaviour over time, which makes them simpler to simulate, these clients are crucial for logistical planning. Understanding repeat behaviour in customer interactions is essential for ERP-based simulations, as it enables organisations to predictably plan delivery routes, frequency, and even inventory allocation (Angolia and Pagliari, 2018). Although those particular routing scenarios were not employed in this study, such consistent demand patterns may be used to model effective delivery routes, prioritisation criteria, or batch scheduling in a SAP context.

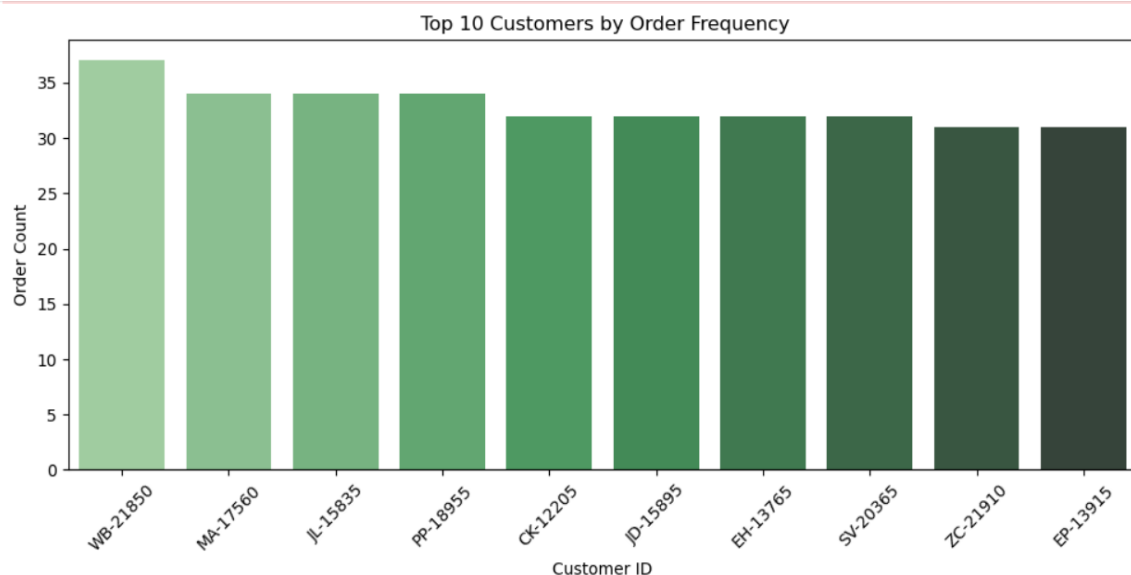


Fig 9. Customer Order Frequency

In emerging economies, where logistical resources are scarce, this idea is fundamental. According to (Chowdhury et al., 2025), frequency-based customer segmentation optimises route planning and lowers delivery costs, which contributes to more efficient delivery systems. Knowing which clients submit orders regularly also helps simulate urban demand clusters and evaluate the system's performance when these clusters unexpectedly grow or decline. The evidence that has been seen also gives support to strategic thinking on client loyalty. Regular clients could anticipate more dependable or quicker service. Even though a priority model is not simulated in this study, identifying these similarities helps future models incorporate delivery criteria tailored to individual customers. (Almasan et al., 2021) argue that the quality of the provided services

might be increased with the help of innovative routing systems, and they will not require additional resources unless they align with the ERP logic, especially when combined with data-driven insights.

6.2 SAP ABAP Simulation Results

6.2.1 Normal vs. Peak Demand Simulation

This simulation is just a hypothetical example of SAP ABAP which can be implemented practically in such way. Using predicted demand, peak-day impacts, and product return scenarios, the ABAP simulation model developed for this study aims to evaluate the following critical logistics performance indicators: delivery time, delivery cost, and the number of vehicles required. The program structure, created using SAP ABAP, processes three datasets to apply the logic: return occurrence flags, peak demand dates, and predicted order quantities. The transaction code used to run the code on ABAP server is Se38. For full source code and technical implementation details, please refer to the Appendix.

6.2.1.1 Logic Flow of the Simulation Code

Each date in the forecast dataset is calculated using a structured loop that forms the basis of the simulation logic. Three major steps have been used to explain the process:

1. Input Data Handling

Three separate CSV files are used as input:

1. `forecast_for_abap.csv`: contains daily forecast values.
2. `peak_days.csv`: contains dates with peak order volumes.
3. `returns_flag.csv`: contains dates with a return occurrence flag (1 = return expected).

These are read into internal tables using FORM routines:

```
read_csv_forecast, read_csv_peak, read_csv_return.
```

Each routine loads and parses the data into structured internal tables, skipping the header and splitting rows using a comma delimiter.

2. Base Calculation per Forecast Entry

For each forecasted day:

1. Delivery time is calculated as:
 $\text{time} = \text{forecast} \times 2$ (every order is assumed to take 2 minutes)
2. Cost is estimated as:
 $\text{cost} = \text{forecast} \times 5$ (each order has a baseline cost of 5 units)
3. Vehicles required is derived from:
 $\text{vehicles} = \text{forecast} / 50$ (each vehicle can handle 50 orders)

This gives a base scenario, assuming normal logistics conditions with no disruption.

3. Adjustment for Returns and Peak Days

Product returns and days with the highest demand are two important operational parameters that are adjusted for in the simulation model. The model applies a cost multiplier of 1.1, increases the delivery time by 15 minutes, and raises the delivery cost by 10% if a return flag is present for a specific date. Similarly, the model adds one extra delivery vehicle, raises the cost by 20% (using a 1.2 multiplier), and increases the delivery time by 30 minutes if the date falls on an assumed peak day. The result is more relevant for logistics planning decision-making due to these conditional adjustments, which enable the simulation to accurately depict the operational disturbances caused by demand spikes and reverse logistics operations.

4. Simulation Output

zsim_lmd_abap				
zsim_lmd_abap				
ABAP Simulation on Retail Dataset forecast				
Date	Forecast	Time	Cost	Vehicles
31-12-2017	12.06	39.12	66.33	0
01-01-2018	12.06	24.12	60.30	0
02-01-2018	12.06	24.12	60.30	0
03-01-2018	12.06	39.12	66.33	0
04-01-2018	12.06	24.12	60.30	0
05-01-2018	12.06	24.12	60.30	0
06-01-2018	12.06	24.12	60.30	0
07-01-2018	12.06	39.12	66.33	0
08-01-2018	12.06	39.12	66.33	0
09-01-2018	12.06	24.12	60.30	0

Fig 10. ABAP Simulation Output

ABAP simulation output on the combined forecasted orders, according to the return events and peak demand days to establish the logistic KPIs. In the dataset, the demand has been forecasted at a steady level of approximately 12.06 units for all days. Delivery costs and delivery times will, however, vary depending on the presence of a return flag. On occasions when the dates do not fall on the same day, such as 31-12-2017, 03-01-2018, 07-01-2018, and 08-01-2018, the return flag is set to 1, which indicates the details of returned deliveries. Consequently, the simulation increases the base delivery time by 15 minutes and the cost by 10 per cent, resulting in 39.12 minutes in time and 66.33 units in cost. The base logic is applied to all other days where there are no returns. In these cases, the delivery time will be 24.12 minutes, and the cost will be 60.30 units. The vehicle needed in all cases reduces to 0 since the units being forecasted are less than the threshold of 50 units that require one complete vehicle. Additionally, no peak days matched, so peak adjustments were not triggered. These outputs help indicate how even slight changes in return activity can influence operations performance in last-mile delivery.

This simulation logic effectively illustrates the impact of dynamic external factors on logistics KPIs, such as peak order days and return activity. It illustrates how real-world data may be incorporated into a basic decision-support model in SAP using ABAP. Using real-time or live data streams in business settings, this approach establishes the conditions for potential expansion.

In addition to this, the simulation was run again with a dummy dataset (used only for validation), to check if the ABAP logic was implemented correctly, especially for handling peak demand and cost adjustments. The output confirms that the model reacts correctly when the input data is meaningful (See figure 11).

<i>zsim_lmd_abap</i>				
zsim_lmd_abap				
ABAP Simulation on Retail Dataset forecast				
Date	Forecast	Time	Cost	Vehicles
20-08-2024	120	270	660	3
21-08-2024	95	190	475	1

Fig 11. ABAP simulation output of dummy dataset

6.2.2 Data Quality Impact

For this research, simulated last-mile delivery performance using three datasets: forecast, peak days, and return flags. The accuracy and utility of the simulation findings were greatly influenced by the quality of these datasets. If the data is not clean, complete, or accurate, the simulation may yield misleading results. For example, if forecast numbers are incorrect or missing, then the delivery time, vehicle requirements, and cost estimates will also be inaccurate. The ABAP simulation logic relies on daily demand values to calculate everything, so even minor errors in this data can result in significant differences in the output. For every date in the prediction forecast data file utilised in the simulation, the same value (12.0591) was repeated. As a result, the simulation did not display any daily change in the basic time, cost, or vehicle utilisation for each day. This made monitoring the model's performance in situations with changing demand challenging. More informative and varied findings from the simulation would be obtained if higher-quality prediction data with realistic, varied values were employed. The binary value (0 or 1) in the return flag dataset made it easy to

identify return scenarios. Delivery time was extended by 15 minutes, and the cost increased by 10% on dates when the return flag was set to 1. This indicates that the return data was organised correctly and that the simulation was clearly affected. However, the adjustments would not apply and the findings would be false if any return data were missing, inaccurate, or did not match the anticipated dates. The predicted data's dates did not correspond with the peak day's dataset. The code's peak adjustment logic was thus never activated during this simulation. In this case, the peak day logic, even if properly implemented, has no effect if the peak dates do not coincide with the anticipated dates. This is a blatant example of data quality mismatch. In simulation models, this emphasises the importance of dataset alignment. More vehicle counts, longer delivery times, and more expenses would have resulted from the peak dates being appropriate. According to research, low-quality data can make predictive models less accurate and potentially lead to incorrect conclusions. In a similar vein, real-time and precise data are essential for obtaining significant outcomes in logistics simulations(WSC : 2018 Winter Simulation Conference : 9-12 December 2018, Gothenburg, Sweden, 2019). Real-world unpredictability and uncertainty will be too much for models to manage if they are trained or evaluated on inaccurate or out-of-date data. Additionally, it has been demonstrated that clean data improves simulation and decision-making accuracy across various sectors, including transportation and logistics (Taratukhin et al., 2023b). This demonstrates that frequent validation, cleaning, and updating of datasets is necessary to improve simulation performance in SAP ABAP or any other platform. In conclusion, solid data forms the framework of reliable simulations. Without it, the model might produce inaccurate results, which could impact urban delivery strategy choices.

7.Limitations

This dissertation effectively used SAP ABAP (for last-mile delivery simulation) and Python (for forecasting and data preparation) to construct a logistics simulation model. Although this is a valuable addition, there are still certain limitations:

1. **Simulation Scope for SAP ABAP:** A simple ABAP application that handled predicted orders, return flags, and days with high demand was used to run the simulation. However, the simulation remained logic-driven and did not expand into full-scale SAP application layers, such as Fiori applications or real-time API calls, due to limitations in the capabilities of the available SAP system (including restricted UI integration and the absence of SAP HANA real-time data services).
2. **Theoretical framework:** The practical implementation of SAP ABAP is just a hypothetical example which can in implemented in practice. The simulation is completely based on scenarios and assumptions to validate.
3. **Absence of SAP ABAP Resources in the Logistics Domain:** One of the significant constraints of this study is the lack of scholarly and practical materials devoted explicitly to the use of SAP ABAP for modelling logistics and last-mile delivery. The majority of existing studies on SAP ABAP are focused either on financial operations, data processing, or regular ERP activities, with minimal attention to using it in the optimisation of supply chains. Due to this, identifying directly relevant references to support or verify the SAP ABAP simulation logic of the project was a challenge. This restriction also limited the possibility of comparing the developed model with real-life logistics applications in SAP settings.
4. **Utilising Openly Accessible Datasets:** A publicly accessible sample dataset, which had been cleaned and altered after preparation, served as the simulation's dataset. Although this aided in the development of a structured model, it constrained the breadth and depth of real-world data elements such as vendor variations, delivery addresses, and payment methods.
5. **Lack of Integration of Real-Time Data:** Despite the study's goal of investigating real-time inputs, technological constraints necessitated the use of static CSV files in the simulation. There was no usage of dynamic traffic streams or live APIs. Due to this, testing the model's performance in unexpected or rapidly changing environments proved challenging.

6. Basic KPI Focus: Delivery time, cost, and vehicle utilisation were the only three KPIs that were the focus of the simulation. Other essential performance measures, such as delivery accuracy, order fill rate, customer experience, and environmental sustainability, were omitted, despite their fundamental logistical importance.

8. Conclusion

This dissertation examined the significance of real-time data and simulation models in enhancing the performance of last-mile delivery (LMD) in emerging economies. A model was created in Python to forecast demand, and SAP ABAP was used to conduct operational tests, with the aim of estimating how delivery-related key performance indicators were influenced by variables such as peak demand days, various return types, and the use of advanced machine tools.

These results showed that the growth in demand or the existence of returns corresponded with the observable increase in delivery cost and time, indicating that more vehicles would be needed. The results showed that incorporating real-time information into simulation environments can provide meaningful insights into logistics decision-making, particularly in uncertain and dynamic urban environments.

As noted, there were relatively few existing LMD simulation models that integrated real-time inputs or considered operational fluctuations. The simulation developed overcame these limitations to some extent, despite not utilising proprietary datasets and being technically constrained by limited access to all the functionality of SAP in use.

Ultimately, the research successfully achieved its objectives by developing a practical yet straightforward model that highlights the importance of real-time data in logistics simulations. Future development may include the use of live data, cloud-based applications (e.g. SAP), or their testing in practical logistics to help improve the accuracy and usefulness of the model.

9. Future Recommendations

Future areas of study on the topic of developing real-time last-mile delivery simulation models in emerging markets should aim to incorporate live data directly into the simulation environment. There is a need to validate the simulation results or similar practices from logistics managers or from organization as there are no such implementation has been done in past. These are influenced by real-time traffic, the active flow of orders, and environmental factors such as weather or roadblocks. Although the existing simulation in SAP ABAP can be described as a structured solution based on historical datasets, integrating real-time input data would enable more dynamic decision-making and scenario planning. A feasible direction to pursue would be the integration of SAP ABAP simulations with SAP Fiori applications or cloud-based dashboards. These dashboards are capable of visualising simulation outputs and allowing real-time performance tracking against logistics KPIs such as time delivery cost efficiency, and vehicle utilisation. This visual mapping will assist logistics planners in making rapid tactical changes to operations based on simulated estimates and real-time operational changes.

Moreover, to increase the model's real-world applicability, future work may involve collaboration with companies that provide logistics or municipal delivery services. This type of cooperation would provide researchers and operators with access to operational data, enabling a comparison of actual performance between simulation and real-life delivery results. This would also contribute to the calibration of simulation parameters and validate the suitability of applying these models in decision support in urban logistics. Data quality is a serious consideration in simulation accuracy. Future studies should involve the automated validation of incoming data, as well as the identification of anomalies to distinguish between consistent and missing data. It should also be noted that diversifying the list of performance indicators, including customer satisfaction, the amount of emissions, or the percentage of successful deliveries, will make simulation output more relevant to both commercial and policy-related applications. As a whole, developing a more dynamic, real-time and contextual simulation framework will lead to a more intelligent last-mile delivery planning and aid in overcoming the peculiarities of logistics in the emerging market environment.

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11. Appendix

Forecast Code

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.api import SimpleExpSmoothing

# Load the Excel file
df = pd.read_excel("Retail-Supply-Chain-Sales-Dataset.xlsx")

# Convert Order Date to datetime
df['Order Date'] = pd.to_datetime(df['Order Date'])

# Group daily order count
daily_orders = df.groupby(df['Order Date'].dt.date).size()

# Forecast next 10 days using Simple Exponential Smoothing
model = SimpleExpSmoothing(daily_orders, initialization_method="heuristic")
fit = model.fit(smoothing_level=0.2, optimized=False)
forecast = fit.forecast(10)

# Plot forecast and last 100 days
plt.figure(figsize=(12, 6))
plt.plot(daily_orders[-100:], label='Actual Orders (Last 100 Days)', color='blue')
plt.plot(pd.date_range(start=daily_orders.index[-1], periods=10, freq='D'), forecast, label='Forecast (Next 10 Days)', linestyle='--', color='orange')
plt.title('Forecast of Daily Orders')
plt.xlabel('Date')
plt.ylabel('Order Count')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

Peak Days Code

```
import seaborn as sns

# Get top 10 days with highest orders
top_peak_days = daily_orders.sort_values(ascending=False).head(10)
top_peak_days_df = top_peak_days.reset_index()
top_peak_days_df.columns = ['Date', 'Order Count']

# Plot
plt.figure(figsize=(10, 5))
sns.barplot(data=top_peak_days_df, x='Date', y='Order Count', palette='Blues_d')
plt.title('Top 10 Peak Sales Days')
plt.xticks(rotation=45)
plt.xlabel('Date')
plt.ylabel('Order Count')
plt.tight_layout()
plt.show()
```

Customer Code

```
# Get top 10 customers
top_customers = df['Customer ID'].value_counts().head(10)
top_customers_df = top_customers.reset_index()
top_customers_df.columns = ['Customer ID', 'Order Count']

# Plot
plt.figure(figsize=(10, 5))
sns.barplot(data=top_customers_df, x='Customer ID', y='Order Count', palette='Greens_d')
plt.title(' Top 10 Customers by Order Frequency')
plt.xticks(rotation=45)
plt.xlabel('Customer ID')
plt.ylabel('Order Count')
plt.tight_layout()
plt.show()
```

Return Code

```
import pandas as pd
import random

# Load forecast data
forecast_df = pd.read_csv("forecast_for_abap.csv")

# Generate Return_Flag column: 30% chance of return
forecast_df['Return_Flag'] = [1 if random.random() < 0.3 else 0 for _ in range(len(forecast_df))]

# Save the required Return Flags for SAP ABAP
forecast_df[['Date', 'Return_Flag']].to_csv('returns_flag.csv', index=False)
```

CSV files used for ABAP Simulation

Forecast

	Date	Forecasted Orders
1	2017-12-31	12.05911977321507
2	2018-01-01	12.05911977321507
3	2018-01-02	12.05911977321507
4	2018-01-03	12.05911977321507
5	2018-01-04	12.05911977321507
6	2018-01-05	12.05911977321507
7	2018-01-06	12.05911977321507
8	2018-01-07	12.05911977321507
9	2018-01-08	12.05911977321507
10	2018-01-09	12.05911977321507

Peak Days

	Date	Order Count
1	2016-05-09	38
2	2017-02-09	36
3	2016-10-11	35
4	2017-01-12	34
5	2017-02-12	34
6	2017-09-12	33
7	2017-08-12	30
8	2017-12-11	30
9	2017-09-09	29
10	2016-11-12	28

Return

	Date	Return_Flag
1	2017-12-31	0
2	2018-01-01	0
3	2018-01-02	0
4	2018-01-03	1
5	2018-01-04	0
6	2018-01-05	0
7	2018-01-06	0
8	2018-01-07	1
9	2018-01-08	1
10	2018-01-09	0

Dummy datasets

Date	Forecast order
20-08-2024	120
21-08-2024	95

Date	Order Count
20-08-2024	40

Date	return_flag
20-08-2024	1

ABAP

Code:

ABAP Editor: Change Report ZSIM_LMD_ABAP

Pattern
Pretty Printer
Text Elements

Report
ZSIM_LMD_ABAP
Active

```

1  REPORT zsim_lmd_abap.
2
3  TYPES: BEGIN OF ty_forecast,
4         date      TYPE d,
5         forecast  TYPE char50,
6     END OF ty_forecast.
7
8  TYPES: BEGIN OF ty_peak,
9         date      TYPE d,
10        order_count TYPE char50,
11    END OF ty_peak.
12
13 TYPES: BEGIN OF ty_return,
14         date      TYPE d,
15         return_flag TYPE char50,
16    END OF ty_return.
17
18 DATA: lt_forecast TYPE TABLE OF ty_forecast,
19        lt_peak     TYPE TABLE OF ty_peak,
20        lt_return   TYPE TABLE OF ty_return,
21        lt_output   TYPE STANDARD TABLE OF string,
22        lv_line     TYPE string.
23
24 * Read CSV into internal tables
25 PERFORM read_csv_forecast USING lt_forecast lv_line CHANGING lt_output

```

No syntax errors found in Report ZSIM_LMD_ABAP

ABAP Editor: Change Report ZSIM_LMD_ABAP

Pattern Pretty Printer Text Elements

Report **ZSIM_LMD_ABAP** Active

```

24  * Read CSV into internal tables
25  PERFORM read_csv_forecast USING 'forecast_for_abap.csv' CHANGING lt_forecast.
26  PERFORM read_csv_peak      USING 'peak_days.csv'          CHANGING lt_peak.
27  PERFORM read_csv_return    USING 'returns_flag.csv'        CHANGING lt_return.
28
29  * Simulate Delivery
30  DATA: lv_cost      TYPE numc5,
31         lv_time      TYPE char8,
32         lv_vehicles  TYPE char50,
33         lv_var1      TYPE p DECIMALS 1 VALUE '1.1',
34         lv_var2      TYPE p DECIMALS 1 VALUE '1.2'.
35
36  LOOP AT lt_forecast INTO DATA(ls_day).
37    DATA(lv_return) = 0.
38    READ TABLE lt_return INTO DATA(ls_ret) WITH KEY date = ls_day-date.
39    IF sy-subrc = 0.
40      lv_return = ls_ret-return_flag.
41    ENDIF.
42
43    * * Basic delivery logic
44    lv_time = ls_day-forecast * 2.
45    lv_cost = ls_day-forecast * 5.
46    lv_vehicles = ls_day-forecast / 50.
47
48
  
```

ABAP

No syntax errors found in Report ZSIM_LMD_ABAP



ABAP Editor: Change Report ZSIM_LMD_ABAP

Pattern Pretty Printer Text Elements

Report **ZSIM_LMD_ABAP** Active

```

47
48
49  * * Adjust for peak load
50  READ TABLE lt_peak INTO DATA(ls_peak) WITH KEY date = ls_day-date.
51  IF sy-subrc = 0.
52    lv_time = lv_time + 30.
53    *   lv_cost = lv_cost * 1.2.
54    lv_cost = lv_cost * lv_var2.
55    lv_vehicles = lv_vehicles + 1.
56  ENDIF.
57
58  * * Adjust for return scenario
59  IF lv_return = 1.
60    lv_time = lv_time + 15.
61    *   lv_cost = lv_cost * 1.1.
62    lv_cost = lv_cost * lv_var1.
63  ENDIF.
64
65  CONCATENATE ls_day-date ls_day-forecast lv_time lv_cost lv_vehicles
66    INTO lv_line SEPARATED BY '|'.
67  APPEND lv_line TO lt_output.
68  ENDLOOP.
69
70  * Display output
71  LOOP AT lt_output INTO lv_line.
  
```

Scope: \LOOP

ABAP

No syntax errors found in Report ZSIM_LMD_ABAP



ABAP Editor: Change Report ZSIM_LMD_ABAP

Report: ZSIM_LMD_ABAP Active

```

70  * Display output
71  LOOP AT lt_output INTO lv_line.
72    WRITE: / lv_line.
73  ENDLOOP.
74
75  *-----*
76  * FORM ROUTINES TO READ CSVs (forecast, peak, return)
77  *-----*
78
79  FORM read_csv_forecast USING p_file TYPE string CHANGING pt_data TYPE STANDARD TABLE.
80    DATA: lv_fullpath TYPE string VALUE 'C:\SAP\forecast_for_abap.csv', " Change as needed
81          lv_line      TYPE string,
82          lt_file       TYPE TABLE OF string,
83          ls_forecast   TYPE ty_forecast.
84
85    CALL METHOD cl_gui_frontend_services=>gui_upload
86      EXPORTING
87        filename = lv_fullpath
88        filetype  = 'CSV'
89      CHANGING
90        data_tab = lt_file.
91
92  LOOP AT lt_file INTO lv_line FROM 2. " Skip header
93    SPLIT lv_line AT ',' INTO ls_forecast-date ls_forecast-forecast.
94    APPEND ls_forecast TO pt_data.
95  ENDLOOP.
96  ENDFORM.

```

Scope: \FORM read_csv_forecast ABAP

✓ No syntax errors found in Report ZSIM_LMD_ABAP

ABAP Editor: Change Report ZSIM_LMD_ABAP

Report: ZSIM_LMD_ABAP Active

```

93    SPLIT lv_line AT ',' INTO ls_forecast-date ls_forecast-forecast.
94    APPEND ls_forecast TO pt_data.
95  ENDLOOP.
96  ENDFORM.
97
98  FORM read_csv_peak USING p_file TYPE string CHANGING pt_data TYPE STANDARD TABLE.
99    DATA: lv_fullpath TYPE string VALUE 'C:\SAP\peak_days.csv', " Change as needed
100          lv_line      TYPE string,
101          lt_file       TYPE TABLE OF string,
102          ls_peak       TYPE ty_peak.
103
104    CALL METHOD cl_gui_frontend_services=>gui_upload
105      EXPORTING
106        filename = lv_fullpath
107      CHANGING
108        data_tab = lt_file.
109
110  LOOP AT lt_file INTO lv_line FROM 2.
111    SPLIT lv_line AT ',' INTO ls_peak-date ls_peak-order_count.
112    APPEND ls_peak TO pt_data.
113  ENDLOOP.
114  ENDFORM.
115
116  FORM read_csv_return USING p_file TYPE string CHANGING pt_data TYPE STANDARD TABLE.
117    DATA: lv_fullpath TYPE string VALUE 'C:\SAP\return_days.csv', " Change as needed
118          lv_line      TYPE string,
119          lt_file       TYPE TABLE OF string,
120          ls_return     TYPE ty_return.
121
122    CALL METHOD cl_gui_frontend_services=>gui_upload
123      EXPORTING
124        filename = lv_fullpath
125      CHANGING
126        data_tab = lt_file.
127
128  LOOP AT lt_file INTO lv_line FROM 2.
129    SPLIT lv_line AT ',' INTO ls_return-date ls_return-order_count.
130    APPEND ls_return TO pt_data.
131  ENDLOOP.
132  ENDFORM.

```

Scope: \FORM read_csv_peak ABAP

✓ No syntax errors found in Report ZSIM_LMD_ABAP

ABAP Editor: Change Report ZSIM_LMD_ABAP

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Pattern Pretty Printer Text Elements

Report ZSIM_LMD_ABAP Active

```
116 FORM read_csv_return USING p_file TYPE string CHANGING pt_data TYPE STANDARD TABLE.
117   DATA: lv_fullpath TYPE string VALUE 'C:\SAP\returns_flag.csv', " Change as needed
118         lv_line      TYPE string,
119         lt_file       TYPE TABLE OF string,
120         ls_ret        TYPE ty_return.
121
122   CALL METHOD cl_gui_frontend_services=>gui_upload
123     EXPORTING
124       filename = lv_fullpath
125     CHANGING
126       data_tab = lt_file.
127
128   LOOP AT lt_file INTO lv_line FROM 2.
129     SPLIT lv_line AT ',' INTO ls_ret-date ls_ret-return_flag.
130     APPEND ls_ret TO pt_data.
131   ENDLOOP.
132 ENDFORM.
```

Ethics Approval Form

5629869 - Ethical approval is granted (WMG Taught Student Research)

Date: **August 29, 2025**

Student: **Priya Rithadia**

Student ID number: **5629869**

Project title: **Optimization of last-mile delivery using Big Data and SAP ERP System**

Your ethical approval number is **WMG-R_2gUvRmeprqqReJl**

Your supervisor/coach **Vasilis Loulos** has **granted ethical approval for your project.**

This means you have consent to conduct your research.

You now have the appropriate approval in place to begin your data collection. It is advisable to note the actual dates of data collection in the final project submission to evidence that your data was collected after the date of ethical approval (as stated in this email).

You are reminded that you must now adhere to the answers and detail given in the completed ethics form. If anything changes in your research such that any of your answers change to the form for which you received **ethical approval** for, then you must contact your supervisor to check if you need to reapply for or update your **ethical approval** before you proceed with data collection.

When you submit your project, please write **your ethical approval number against the ethical approval field** on the cover page of the submission and include a copy of this email in the Appendices of the submission.

Kind regards,

WMG Projects Team

[Download as PDF](#)